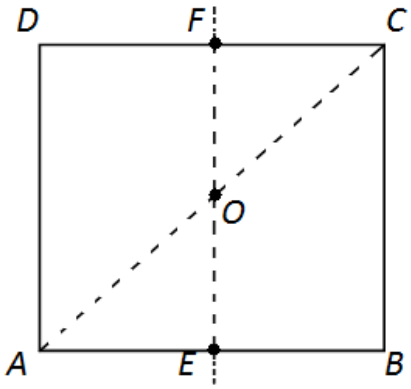
Q1

For the given uniform square lamina $ABCD$, whose centre is O,



(1) $I_{\mathrm{AC}} = \sqrt{2} I_{\mathrm{EF}}$

(2) $\sqrt{2} I_{\mathrm{AC}} = I_{\mathrm{EF}}$

(3) $I_{\mathrm{AD}} = 3 I_{\mathrm{EF}}$

(4) $I_{\mathrm{AC}} = I_{\mathrm{EF}}$

Q2

Assertion : In an astronomical telescope, the objective lens has more aperture diameter

Reason : If we increase the focus distance of the objective lens, the magnification power of the telescope increases

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

Q3

A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of 1 m from the diode is

(1) $7.75\ \mathrm{V\ m^{-1}}$

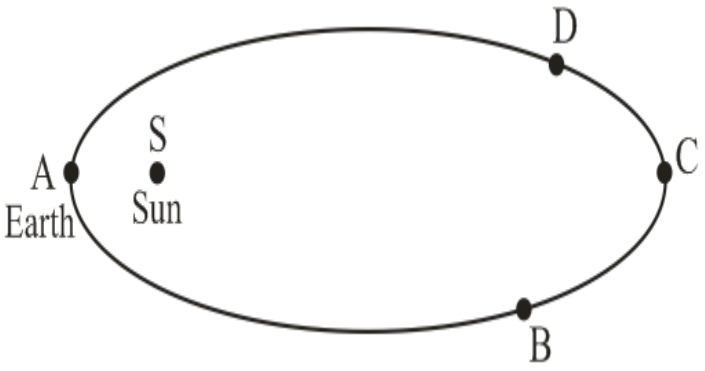
(2) $1.73\ \mathrm{V\ m^{-1}}$

(3) $2.45\ \mathrm{V\ m^{-1}}$

(4) $5.48\ \mathrm{V\ m^{-1}}$

Q4

The earth moves in an elliptical orbit with the sun S at one of the foci as shown in the figure. Its kinetic energy is maximum at the point



(1) A

(2) B

(3) C

(4) D

Q5

In a hypothetical Bohr hydrogen, the mass of the electron is doubled. The energy E_0 and radius r_0 of the first orbit will be (a_0 is the Bohr's first orbit radius of hydrogen)

(1) $E_0 = -27.2$ eV, $r_0 = \frac{a_0}{2}$

(2) $E_0 = -27.2$ eV, $r_0 = a_0$

(3) $E_0 = -13.6$ eV, $r_0 = \frac{a_0}{2}$

(4) $E_0 = -13.6$ eV, $r_0 = a_0$

Q6

The dimensional formula of the constant *a* in Vander Waal's gas equation $\left(P + \frac{a}{V^2}\right)(V - b) = RT$ is:

(1) $[ML^4T^{-1}]$

(2) $[ML^2T^{-2}]$

(3) $[ML^5T^{-3}]$

(4) $[ML^5T^{-2}]$

Q7

In a YDSE : distance between slits and screen is $D = 1$ m, separation between the slits is d $= 1$ mm and wavelength of the incident light is $\lambda = 5000$ nm. The distance of 100th maxima from the central maxima is:

(1) $\frac{1}{2}$ m

(2) $\frac{\sqrt{3}}{2}$ m

(3) $\frac{1}{\sqrt{3}}$ m

(4) does not exist.

Q8

One thousand small water drops of equal size combine to form a big drop. The ratio of the final surface energy to the total initial surface energy is

(1) 1 : 10

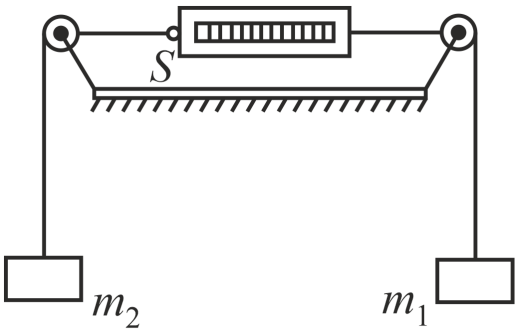
(2) 10 : 1

(3) 1000 : 1

(4) 1 : 1000

Q9

Find the reading S which is a massless spring balance. In the arrangement shown, the pulleys are fixed and ideal, the springs are light and $m_1 > m_2$.(in units of mass)



(1) m_1 - m_2

(2) $\frac{1}{2}(m_1 + m_2)$

(3) $\frac{m_1 m_2}{m_1+m_2}$

(4) $\frac{2m_1 m_2}{m_1+m_2}$

Q10

The relation between Y, η and B is:

Where Y is Young's modulus, B is Bulk modulus and η is modulus of rigidity.

(1) $\frac{1}{Y} = \frac{1}{3\eta} + \frac{1}{9B}$

(2) $\frac{9}{Y} = \frac{1}{\eta} + \frac{3}{B}$

(3) $\frac{1}{\eta} = \frac{1}{B} + \frac{1}{Y}$

(4) $\frac{9}{Y} = \frac{3}{\eta} + \frac{1}{B}$

Q11

A plano-convex lens fits exactly into a plano-concave lens. Their plane surfaces are parallel to each other. If lenses are made of different materials of refractive indices μ_1 and μ_2, R is the radius of curvature of the curved surface of the lenses, then the focal length of combination is

(1) $\frac{R}{2(\mu_1+\mu_2)}$

(2) $\frac{R}{2(\mu_1-\mu_2)}$

(3) $\frac{R}{(\mu_1-\mu_2)}$

(4) $\frac{2R}{\mu_1-\mu_2}$

Q12

A tunnel is made across the earth of radius R, passing through its centre. A ball is dropped from a height h in the tunnel. The motion will be periodic with time period.

(1) $2\pi\sqrt{\frac{R}{g}}+4\sqrt{\frac{h}{g}}$

(2) $2\pi\sqrt{\frac{R}{g}}+4\sqrt{\frac{2h}{g}}$

(3) $2\pi\sqrt{\frac{R}{g}}+\sqrt{\frac{h}{g}}$

(4) $2\pi\sqrt{\frac{R}{g}}+\sqrt{\frac{2h}{g}}$

Q13

A 20 μF capacitor is charged by a constant current of 10 mA . If the capacitor is initially uncharged how much time does it take for the potential difference across the capacitor to reach 300 V .

(1) 0.6 sec

(2) 0.8 sec

(3) 1 sec

(4) 0.4 sec

Q14

The potential gradient of a potentiometer is $0.2\ \mathrm{V\ m^{-1}}$. A current of $0.1\ \mathrm{A}$ is flowing through a coil of resistance $2\ \Omega$. The balancing length (in meters) for the potential difference at the ends of this coil will be,

(1) $2\ \mathrm{m}$

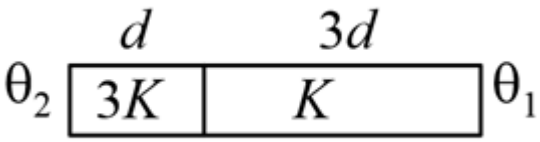
(2) $1\ \mathrm{m}$

(3) $0.2\ \mathrm{m}$

(4) $0.1\ \mathrm{m}$

Q15

Two materials having coefficient of thermal conductivity 3k and k and thickness d and 3d respectively, are joined to form a slab as shown in the figure. The temperature of the outer surfaces are θ_2 and θ_1 and $(\theta_2 > \theta_1)$, Then the temperature at the interface is:



(1) $\frac{\theta_1}{6} + \frac{5\theta_2}{6}$

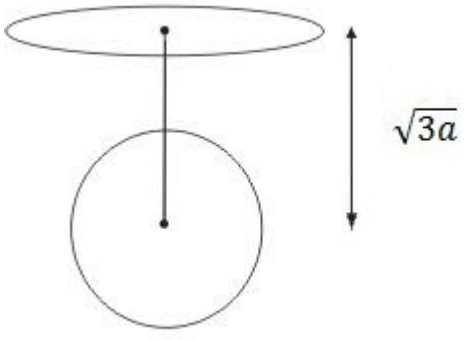
(2) $\frac{\theta_1}{3} + \frac{2\theta_2}{3}$

(3) $\frac{\theta_2+\theta_1}{2}$

(4) $\frac{\theta_1}{10} + \frac{9\theta_2}{10}$

Q16

A uniform ring of mass m and radius a is placed directly above a uniform sphere of mass M and of equal to radius of ring. The centre of the ring is at a distance $\sqrt{3}$a from the centre of the sphere as shown in figure . The gravitational force (F) exerted by the sphere on the ring is



(1) $\frac{\sqrt{3}\,GMm}{8a^2}$

(2) $\frac{2\,GMm}{3a^2}$

(3) $\frac{7\ \mathrm{GMm}}{\sqrt{2}\ \mathrm{a}^2}$

(4) $\frac{3\ \mathrm{GMm}}{\mathrm{a}^2}$

Q17

Assertion: For looping a vertical loop of radius r, the minimum velocity at the lowest point should be $\sqrt{5\,\mathrm{gr}}$

Reason: Velocity at the highest point would be zero.

(1) Assertion is True, Reason is True; Reason is a correct explanation for Assertion.

(2) Assertion is True, Reason is True; Reason is not a correct explanation for Assertion.

(3) Assertion is True, Reason is False.

(4) Assertion is False but Reason is True.

Q18

A block of mass 0.50 kg is moving with a speed of 2.00 ms^{-1} on a smooth surface. It strikes another mass of 1.00 kg and then they move together as a single body. The energy loss during the collision is

(1) 0.34 J

(2) 0.16 J

(3) 1.00 J

(4) 0.67 J

Q19

In an ac circuit, an alternating voltage $e = 200\sqrt{2}\ \sin 100t$ volts is connected to a capacitor of capacity $1\ \mu\mathrm{F}$. The rms value of the current in the circuit is

(1) $10\ \mathrm{mA}$

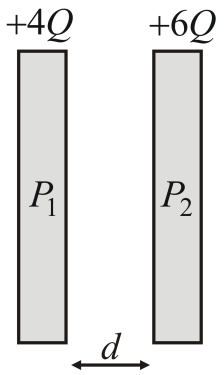
(2) $100\ \mathrm{mA}$

(3) $200\ \mathrm{mA}$

(4) $20\ \mathrm{mA}$

Q20

Two identical conducting very large plates P_1 and P_2 having charges $+4Q$ and $+6\ Q$ are placed very closed to each other at separation d. The plate area of either face of the plate is A. The potential difference between plates P_1 and P_2 is

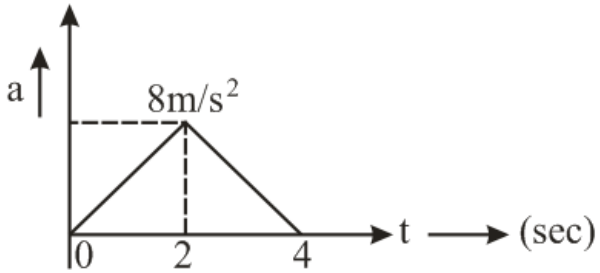


In the above question, if plates P_1 and P_2 are connected by a thin conducting wire, then the amount of heat produced will be

(1) $\frac{Q^2}{A\varepsilon_0}d$
(2) $\frac{5Q^2}{A\varepsilon_0}d$
(3) $\frac{2Q^2}{A\varepsilon_0}d$
(4) None of these

Q21

Acceleration time graph of a particle moving in a straight line is shown. Velocity of particle at $t = 0$ is 5 m s^{-1}. The velocity (in m s^{-1})of the particle at the end of fourth second is



Q22

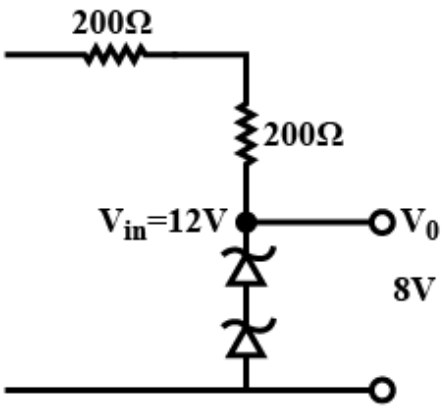
The value of inductance L for which the current is maximum in a series LCR AC circuit with $C = 10\ \mu\text{F}$ and $\omega = 1000\ \text{s}^{-1}$ is $(20 \times n)\ \text{mH}$, $n =$?

Q23

The density of a sphere is measured by measuring the mass and diameter. If it is known that the maximum percentage errors in the measurement of mass and diameter are 2% and 3% respectively then the maximum percentage error in the measurement of density is

Q24

In the circuit shown below, is working as a 8 V dc regulated voltage source. When 12 V is used as an input, the power dissipated (in mW) in each diode is (Considering both zener diodes are identical)



Q25

A body cools from 80°C to 50°C in 5 min. Calculate the time it takes to cool from 60°C to 30°C. The temperature of the surroundings is 20°C.

Q26

Find the closest integer wavelength value of K_α line (in picometer) in copper (Z = 29) if the wavelength of K_α = line in iron (Z = 26) is known to be equal to193 picometer.

Q27

The speed of sound in air is $350\ \text{m s}^{-1}$. The fundamental frequency of an open pipe $50\ \text{cm}$ long will be a Hz, what is the value of a?

Q28

Gas at pressure P_0 is contained in a vessel. If the mass of all the molecules are halved and their speed is doubled then the resulting pressure becomes nP_0. What is the value of ' n'.

Q29

A TV tower has a height of 400 m at a given place. Its coverage area is nearly $1.6 \times 10^{\text{x}}\ \text{m}^2$, then fill x (Take radius of earth =6400 km)

Q30

An electric field is given by $\vec{E} = \left(y\hat{\text{i}} + x\hat{\text{j}}\right)\ \text{N C}^{-1}$. Find the work done (in J) in moving a 1 C charge from $\vec{r}_{\text{A}} = \left(2\hat{\text{i}} + 2\hat{\text{j}}\right)\ \text{m}$ to $\vec{r}_{\text{B}} = \left(4\hat{\text{i}} + \hat{\text{j}}\right)\ \text{m}$

Q31

Assertion: PbO_2 is an oxidising agent and reduced to PbO.

Reason: Stability of $\text{Pb(II)} > \text{Pb(IV)}$ on account of inert pair effect.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

Q32

Read the following statements about enzymes. Which among them are correct?

I. The mechanism by which enzymes work is known as lock and key.

II. They denature at high temperatures.

III. They are mostly proteins having a tertiary structure.

IV. They may be assisted by vitamins and minerals.

(1) I, II and III only

(2) I, II, III and IV

(3) II and IV only

(4) I, II and IV only

Q33

Given below are two statements:

Statement I : None of the alkaline earth metal hydroxides dissolve in alkali.

Srtatement II : Solubility of alkaline earth metal hydroxides in water increases down the group.

In the light of the above statements, choose the most appropriate answer from the options given below:

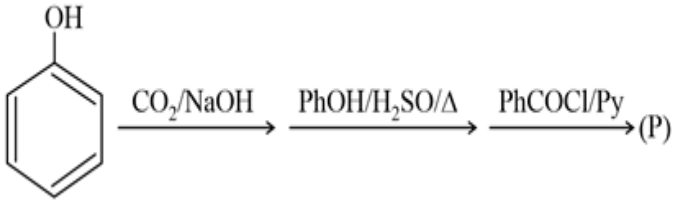
(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

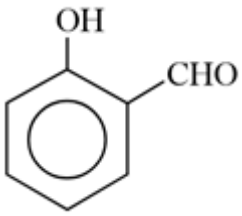
(3) Statement I and Statement II both are incorrect.

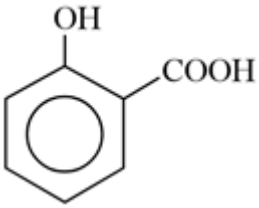
(4) Statement I and Statement II both are correct.

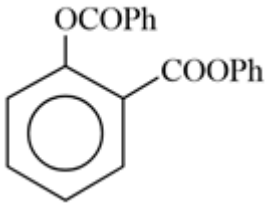
Q34

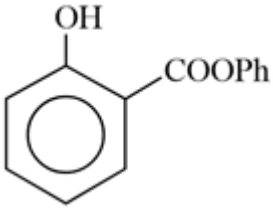


Give the product (P) of the above reaction sequence.

(1) 

(2) 

(3) 

(4) 

Q35

You can do a simple demonstration at home of the effects of osmosis on a biological cell: Immerse two chicken eggs in vinegar for a few days to dissolve the eggshells and expose the membranes, then place one egg in pure water and the other in syrup for a couple more days. Based on what you know about osmosis, what would you predict will happen to the eggs ?

(1) Both eggs will shrink .

(2) The egg in water will shrink, while the egg in syrup will swell .

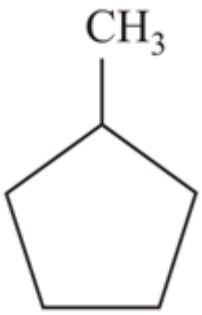
(3) Both eggs will swell .

(4) The egg in water will swell, while the egg in syrup will shrink .

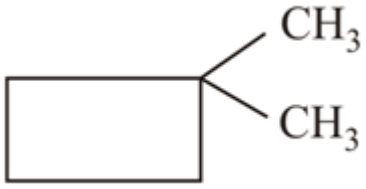
Q36

Which of the following cycloalkanes will show cis-trans isomerism?

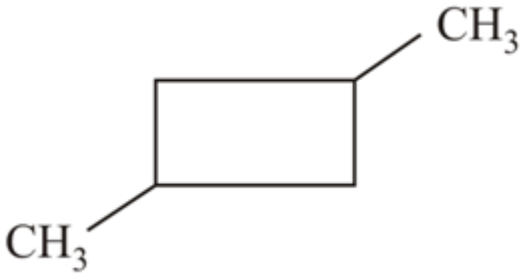
(a)

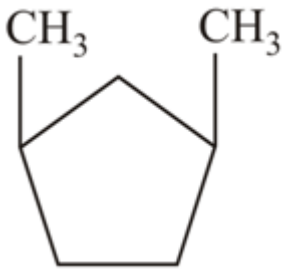


(b)



(c)





(d)

(1) Only (a)

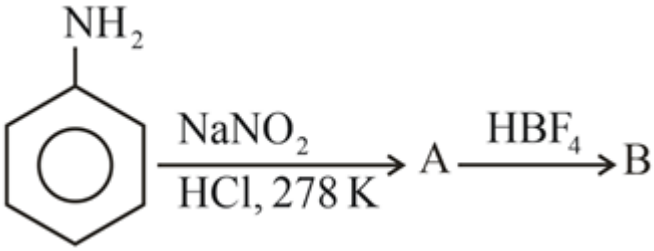
(2) Only (b)

(3) Both (c) and (d)

(4) Both (a) and (b)

Q37

In the chemical reactions,



The compounds A and B, respectively, are

(1) nitrobenzene and fluorobenzene

(2) phenol and benzene

(3) benzene diazonium chloride and fluorobenzene

(4) nitrobenzene and chlorobenzene

Q38

The standard emf of the cell $\mathrm{Zn}\left|\mathrm{Zn}^{2+}\right\|\mathrm{Ag}^{+}\left|\mathrm{Ag}\right.$ is 1.56 V. If the standard reduction potential of Ag is $0.8V$, the standard oxidation potential of Zn is

(1) -0.76 V

(2) $+0.76$ V

(3) -2.36 V

(4) $+2.36$ V

Q39

The chemical substances that raise the pH value of the stomach are called as _____.

(1) antacids

(2) antipyretics

(3) analgesics

(4) antibotics

Q40

Match the catalysts to the correct process.

	Catalyst		**Process**
A.	$\mathrm{TiCl_3}$	**i.**	Wacker process
B.	$\mathrm{PdCl_2}$	**ii.**	Ziegler – Natta polymerisation
C.	$\mathrm{CuCl_2}$	**iii.**	Contact process`
D.	$\mathrm{V_2O_5}$	**iv.**	Deacon's process

(1) A $\rightarrow$ iii , B $\rightarrow$ i , C $\rightarrow$ ii , D $\rightarrow$ iv

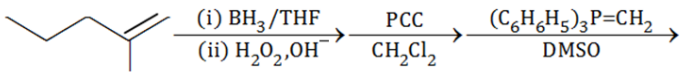
(2) A $\rightarrow$ iii , B $\rightarrow$ ii , C $\rightarrow$ iv , D $\rightarrow$ i

(3) A $\rightarrow$ ii , B $\rightarrow$ i , C $\rightarrow$ iv , D $\rightarrow$ iii

(4) A $\rightarrow$ ii , B $\rightarrow$ iii , C $\rightarrow$ iv , D $\rightarrow$ i

Q41

What is the product of the reaction sequence below?



(1) 2-methyl-1-hexene

(2) 2,3-dimethyl-2-pentene

(3) 2-methyl-2-hexene

(4) 3-methyl-1-hexene

Q42

Consider the reactions

(A) $H_2O_2 + 2HI \rightarrow I_2 + 2H_2O$

(B) $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$

Which of the following statements is correct about H_2O_2 with reference to these reactions? Hydrogen peroxide is _____.

(1) an oxidising agent in both (A) and (B)

(2) an oxidising agent in (A) and reducing agent in (B)

(3) a reducing agent in (A) and oxidising agent in (B)

(4) a reducing agent in both (A) and (B)

Q43

The pollution by _____ causes minamata disease.

(1) Organic waste into drinking water

(2) Oil spill in water

(3) Industrial waste mercury into fishing water

(4) Arsenic into the atmosphere

Q44

Given below are two statements:

Statement I: A mixture of chloroform and aniline can be separated by simple distillation.

Statement II: When separating aniline from a mixture of aniline and water by steam distillation aniline boils below its boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are true

(2) Statement I is true but Statement II is false

(3) Statement I is false but Statement II is true

(4) Both Statement I and Statement II are false

Q45

The activation energies of two reactions are E_1 and E_2 $(E_1 > E_2)$. If the temperature of the system is increased from T_1 to T_2, the rate constants of the reactions change from k_1 to k_1' in the first reaction and k_2 and k_2' in the second reaction. Predict which of the following expressions is correct.

(1) $\frac{k_1'}{k_1'} = \frac{k_2'}{k_2'}$

(2) $\frac{k_1'}{k_1} > \frac{k_2'}{k_2}$

(3) $\frac{k_1'}{k_1'} < \frac{k_2'}{k_2'}$

(4) $\frac{k_1'}{k_1'} = \frac{k_2'}{k_2'} = 1$

Q46

In extractive metallurgy of zinc, partial fusion of ZnO with coke is called sintering and reduction of the ore to the molten metal is called

(1) Smelting

(2) Calcining

(3) Roasting

(4) Sintering

Q47

The oxidation states of 'P' in $H_4P_2O_7$, $H_4P_2O_5$ and $H_4P_2O_6$, respectively, are :

(1) $7,\ 5$ and 6

(2) $5,\ 4$ and 3

(3) $5,\ 3$ and 4

(4) $6,\ 4$ and 5

Q48

Among the following statements, which statements are correct if O_2 changes to O_2^+?

(A) Magnetic moment decreases.

(B) Bond length increases.

(C) Bond order increases.

(D) Magnetic moment remains the same.

(1) A and B

(2) B and D

(3) A and C

(4) A and D

Q49

Heat of formation of H_2O is -188 kJ/mol and H_2O_2 is -286 kJ/mol. The enthalpy change for the reaction $2H_2O_2 \rightarrow 2H_2O + O_2$ is

(1) 196 kJ

(2) -196 kJ

(3) 984 kJ

(4) -984 kJ

Q50

Nylon-66 is made up of monomers hexamethylene diamine and _____.

(1) Sulphurhexa fluoride

(2) Adipic acid

(3) Sulphurous acid

(4) Phthalic acid

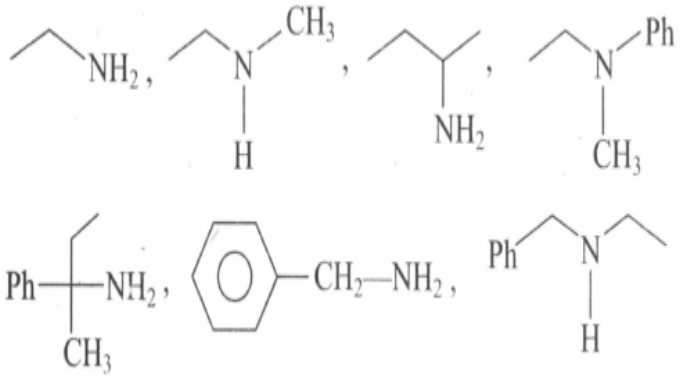
Q51

What is the % dissociation of H_2S ? If one mole of H_2S is introduced in 1 litre vessel at 100 K if K_c for the reaction

$2H_2S(g) \rightleftharpoons 2H_2(g) + S_2(g)$ is 4 x10^{-6} (Assume negligible dissociation to approximate answer to nearest integer)

Q52

Of the following amines how many can be separated by Hoffmann's mustard oil reaction?



Q53

When $35\ \text{mL}$ of $0.15\ \text{M}$ lead nitrate solution is mixed with $20\ \text{mL}$ of $0.12\ \text{M}$ chromic sulphate solution, N $\times 10^{-5}$ moles of lead sulphate precipitate out. Find integral value of N.

Q54

If the freezing point of a 0.01 molal aqueous solution of a cobalt (III) bromide-ammonia complex (which behaves as a strong electrolyte) is -0.0557°C, the number of bromide(s) in the coordination sphere of the complex is [K_f for water $= 1.86\ \text{K kg mol}^{-1}$]

Q55

The number of optical isomers possible for $\left[\text{Cr}\left(\text{C}_2\text{O}_4\right)_3\right]^{3-}$ is

Q56

A solution is obtained by mixing 200 g of 30% and 300 g of 20% solution by weight. What is the percentage of solute in the final solution?

Q57

The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when light of 300 nm wavelength falls on the metal is :

Metal	Li	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ (eV)	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

Q58

If the equilibrium constant of the reaction of weak acid HA with strong base is 10^9 , then the pH of 0.1 M NaA solution will be :

Q59

KBr is doped with 10^{-5} mole percent of $SrBr_2$ The number of cationic vacancies in 1 g of KBr crystal is N $\times 10^{14}$. Find integral value of N. [Atomic Mass : $\mathrm{K}=39.1 \mathrm{u}, \mathrm{Br}=79.9 \mathrm{u}, \mathrm{N}_{\mathrm{A}}=6.023 \times 10^{23}$]

Q60

The inversion of Cane sugar proceeds with constant half-life of 500 minutes at $pH = 5$ for any concentration of sugar. However if pH $= 6$, the half-life changes to 50 minutes. For this reaction rate equation is rate $= k[\text{ sugar }]^m[H^+]^n$.

The value of " n " is.

Q61

Let $a > 1$ be a real number and $f(x) = \log_a x^2$ for $x > 0$. If f^{-1} is the inverse function of f, and b and c are real numbers, then $f^{-1}(b + c)$ is equal to

(1) $f^{-1}(b) \cdot f^{-1}(c)$

(2) $f^{-1}(b) + f^{-1}(c)$

(3) $\frac{1}{f(b+c)}$

(4) $\frac{1}{f^{-1}(b)+f^{-1}(c)}$

Q62

Let $\overrightarrow{a}=\hat{\mathrm{i}}+2\hat{\mathrm{j}}-3\widehat{\mathrm{k}}$ and $\overrightarrow{b}=2\hat{\mathrm{i}}-3\hat{\mathrm{j}}+5\widehat{\mathrm{k}}$. If $\overrightarrow{r}\times\overrightarrow{a}=\overrightarrow{b}\times\overrightarrow{r}$, $\overrightarrow{r}\cdot\left(\alpha\hat{\mathrm{i}}+2\hat{\mathrm{j}}+\widehat{\mathrm{k}}\right)=3$ and $\overrightarrow{r}\cdot\left(2\hat{\mathrm{i}}+5\hat{\mathrm{j}}-\alpha\widehat{\mathrm{k}}\right)=-1, \alpha\in R$, then the value of $\alpha+\left|\overrightarrow{r}\right|^2$ is equal to :

(1) 9

(2) 15

(3) 13

(4) 11

Q63

A ray of light passes through the points $A(2,\ 3)$ reflected at a point B on the line $x+y=0$ and then, passes through $(5,\ 3)$. Then the coordinates of B are

(1) $\left(\frac{1}{3},\ -\frac{1}{3}\right)$

(2) $\left(\frac{2}{5},\ -\frac{2}{5}\right)$

(3) $\left(\frac{1}{13},\ -\frac{1}{13}\right)$

(4) None of these

Q64

For what value of λ,the system of equations $x+y+z=6,\ \ x+2y+3z=10,\ \ x+2y+\lambda z=10$ is consistent?

(1) 1

(2) 2

(3) -1

(4) 3

Q65

Sum of series of $\cot^{-1}\frac{5}{\sqrt{3}}+\cot^{-1}\frac{9}{\sqrt{3}}+\cot^{-1}\frac{15}{\sqrt{3}}+\cot^{-1}\frac{23}{\sqrt{3}}+\ldots\ldots\infty$ is equal to

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{6}$

(4) $\frac{\pi}{12}$

Q66

If the mirror image of the point $(1, 3, 5)$ with respect to the plane $4x - 5y + 2z = 8$ is (α, β, γ), then $5(\alpha + \beta + \gamma)$ equals :

(1) 43

(2) 47

(3) 41

(4) 39

Q67

The number of 3×3 matrices M with entries from $\{0,1, 2\}$, such that the sum of the diagonal elements of $M^T M$ is 5, are

(1) 198

(2) 126

(3) 135

(4) 162

Q68

The contrapositive of the statement: "If the weather is fine then my friends will come and we will go for a picnic." is

(1) The weather is fine but my friends will not come or we do not go for a picnic.

(2) If my friends do not come or we do not go for picnic then weather will not be fine.

(3) If the weather is not fine then my friends will not come or we do not go for a picnic.

(4) The weather is not fine but my friends will come and we go for a picnic.

Q69

Throwing a biased die, a person will get 5 Rupees if he throws the number 5 and will get 8 Rupees for any other number, then the expected income (in Rupees) per throw is (it is given that the number 5 will appear 5 times as frequently as any other number)

(1) 6.5

(2) 7

(3) 5

(4) 5.5

Q70

The coefficient of x^n in the polynomial $\left(x+{}^{2n+1}C_0\right)\left(x+{}^{2n+1}C_1\right)\left(x+{}^{2n+1}C_2\right)$ ..... $\left(x+{}^{2n+1}C_n\right)$ is

(1) 2^{n+1}

(2) $2^{2n+1}-1$

(3) $2^{2n}-1$

(4) 2^{2n}

Q71

On the set N of all natural numbers define the relation R by aRb if and only if the GCD of a and b is 2, then R is

(1) Reflexive, but not symmetric

(2) Symmetric only

(3) Reflexive, and transitive

(4) Reflexive, symmetric and transitive

Q72

Find the sum of the series $1^2+2^2-3^2+4^2+5^2-6^2+7^2+8^2-9^2+\ldots+$ upto (30) terms

(1) 2425

(2) 2525

(3) 2524

(4) 2562

Q73

If $z \in C$ and satisfies the equation $(z-\bar{z})^2 = 4|z|^2 - 12$, then the maximum value of $|z|$ is

(1) $\sqrt{6}$

(2) $2\sqrt{3}$

(3) $\sqrt{3}$

(4) $\frac{\sqrt{3}}{2}$

Q74

Let AB be the chord of contact of the point $(5, -5)$ w.r.t the circle $x^2 + y^2 = 5$, then the locus of the orthocentre of ΔPAB where P is any point moving on the circle is

(1) $(x-1)^2 + (y+1)^2 = 5$

(2) $\left(x-1\right)^2 + \left(y+1\right)^2 = \frac{5}{2}$

(3) $(x+1)^2 + (y-1)^2 = 5$

(4) $\left(x+1\right)^2 + \left(y-1\right)^2 = \frac{5}{2}$

Q75

A positive integer n is of the form $n = 2^{\alpha}3^{\beta}$, where $\alpha \geq 1$, $\beta \geq 1$. If n has 12 positive divisors and $2n$ has 15 positive divisors, then the number of positive divisors of $3n$ is

(1) 15

(2) 16

(3) 18

(4) 20

Q76

The sum of the roots of the equation $2^{(33x-2)} + 2^{(11x+2)} = 2^{(22x+1)} + 1$ is

(1) $\frac{1}{11}$

(2) $\frac{2}{11}$

(3) $\frac{3}{11}$

(4) $\frac{4}{11}$

Q77

Let S be the set of real values of λ for which the function $f(x) = x^3 - 3(2\lambda - 1)x^2 + 6\lambda x$ has exactly one local maximum and exactly one local minimum, then S can be

(1) $(0,\ 6)$

(2) $(1,\ 4)$

(3) $(-\infty,\ 0)$

(4) $(0,\ \infty)$

Q78

The general solution of the differential equation, $\sin\ 2x\left(\frac{dy}{dx} - \sqrt{\tan\ x}\right) - y = 0$, is

(1) $y\sqrt{\tan\ x} = x + c$

(2) $y\sqrt{\cot\ x} = \tan\ x + c$

(3) $y\sqrt{\tan\ x} = \cot\ x + c$

(4) $y\sqrt{\cot\ x} = x + c$

Q79

For each $t \in R$, let $[t]$ be the greatest integer less than or equal to t. Then $\lim\limits_{x \to 0^+} x\left(\left[\frac{1}{x}\right] + \left[\frac{2}{x}\right] + \ldots + \left[\frac{15}{x}\right]\right)$

(1) does not exist $(\text{in}\ R)$

(2) is equal to 0

(3) is equal to 15

(4) is equal to 120

Q80

A straight line PQ touches ellipse $\frac{x^2}{(3)^2}+\frac{y^2}{(1)^2}=1$ and $circle\ x^2+y^2=4$. RS is a focal chord of ellipse. If RS is parallel to PQ and RS meets the circle at points R' and S', then the length of $R'S'$ is

(1) 1 unit

(2) 2 unit

(3) 3 unit

(4) 4 unit

Q81

If the area bounded by the curve $y=\cos^{-1}(\cos x)$ and $y=|x-\pi|$ is $\frac{\pi^2}{n}$, then n is equal to

Q82

Let z and w be two complex numbers such that $w=z\bar{z}-2z+2$, $\left|\frac{z+i}{z-3i}\right|=1$ and $\mathrm{Re}(w)$ has minimum value. Then, the minimum value of $n\in N$ for which w^n is real, is equal to _____.

Q83

Let a, b, c be positive integers such that $\frac{b}{a}$ is an integer. If a, b, c are in geometric progression and the arithmetic mean of a, b, c is b + 2, then the value of $\frac{a^2+a-14}{a+1}$ is _____

Q84

If 2 distinct numbers are between 0 to 180 (both inclusive) and the probability that their average is 60 is k, then $1086k$ is equal to

Q85

Consider two vectors $\vec{a} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{j} + 2\hat{k}$. If $\vec{c}$ is a unit vector and k be the maximum value of $a \cdot \left(\vec{b} \times \vec{c}\right)$, then the value of $k^2 - 50$ is equal to

Q86

Let $P = \begin{bmatrix} -30 & 20 & 56 \\ 90 & 140 & 112 \\ 120 & 60 & 14 \end{bmatrix}$ and $A = \begin{bmatrix} 2 & 7 & \omega^2 \\ -1 & -\omega & 1 \\ 0 & -\omega & -\omega + 1 \end{bmatrix}$ where $\omega = \frac{-1+i\sqrt{3}}{2}$, and I_3 be the identity matrix of order 3. If the determinant of the matrix $\left(P^{-1}AP - I_3\right)^2$ is $\alpha\omega^2$, then the value of α is equal to _____.

Q87

A sample of 20 observations has mean of 50 and variance of 1, while a sample of 40 observations has mean of 50 and standard deviation 2. The 2 samples are combined to give complete set of 60 observations with variance σ^2, then $3\sigma^2$ is equal to

Q88

Let P be a point in the first octant, whose image Q in the plane $x + y = 3$ (that is, the line segment PQ is perpendicular to the plane $x + y = 3$ and the mid-point of PQ lies in the plane $x + y = 3$) lies on the z-axis. Let the distance of P from the x-axis be 5. If R is the image of P in the xy-plane, then the length of PR is _____.

Q89

If $C_r \equiv {}^{25}C_r$ and $C_0 + 5 \bullet C_1 + 9 \bullet C_2 + \ldots + (101) \bullet C_{25} = 2^{25} \bullet k$, then k is equal to _____.

Q90

Let $f: R \to R$ is a function defined as $f(x) = \begin{cases} |x - [x]| & : [x] \text{ is odd} \\ |x - [x+1]| & : [x] \text{ is even} \end{cases}$, where [.] denotes the greatest integer function, then $\int_{-2}^{4} f(x)\, dx$ is equal to

Answer Key

Q1 (4)	**Q2** (2)	**Q3** (3)	**Q4** (1)
Q5 (1)	**Q6** (4)	**Q7** (3)	**Q8** (1)
Q9 (4)	**Q10** (4)	**Q11** (3)	**Q12** (2)
Q13 (1)	**Q14** (2)	**Q15** (4)	**Q16** (1)
Q17 (3)	**Q18** (4)	**Q19** (4)	**Q20** (4)
Q21 (21)	**Q22** (5)	**Q23** (11)	**Q24** (40)
Q25 (9)	**Q26** (154)	**Q27** (350)	**Q28** (2)
Q29 (10)	**Q30** (0)	**Q31** (1)	**Q32** (2)
Q33 (2)	**Q34** (3)	**Q35** (4)	**Q36** (3)
Q37 (3)	**Q38** (2)	**Q39** (1)	**Q40** (3)
Q41 (4)	**Q42** (2)	**Q43** (3)	**Q44** (1)
Q45 (2)	**Q46** (1)	**Q47** (3)	**Q48** (3)
Q49 (1)	**Q50** (2)	**Q51** (2)	**Q52** (4)
Q53 (525)	**Q54** (1)	**Q55** (2)	**Q56** (24)
Q57 (4)	**Q58** (9)	**Q59** (5)	**Q60** (0)
Q61 (1)	**Q62** (2)	**Q63** (3)	**Q64** (4)
Q65 (2)	**Q66** (2)	**Q67** (1)	**Q68** (2)
Q69 (1)	**Q70** (4)	**Q71** (2)	**Q72** (2)

Q73 (3)
Q74 (1)
Q75 (2)
Q76 (2)

Q77 (3)
Q78 (4)
Q79 (4)
Q80 (2)

Q81 (2)
Q82 (4)
Q83 (4)
Q84 (4)

Q85 (59)
Q86 (36)
Q87 (9)
Q88 (8)

Q89 (51)
Q90 (3)

Q1

By perpendicular axes theorem,

$I_{\mathrm{EF}}=M\frac{a^2+b^2}{12}=\frac{M\left(a^2+a^2\right)}{12}=M\frac{2a^2}{12}$

$I_{\mathrm{z}}=\frac{M\left(2a^2\right)}{12}+\frac{M\left(2a^2\right)}{12}=\frac{Ma^2}{3}.$

By perpendicular axes theorem,

$I_{\mathrm{AC}}+I_{\mathrm{BD}}=I_{\mathrm{z}}\Rightarrow I_{\mathrm{AC}}=\frac{I_{\mathrm{z}}}{2}=\frac{Ma^2}{6}.$

By the same theorem,

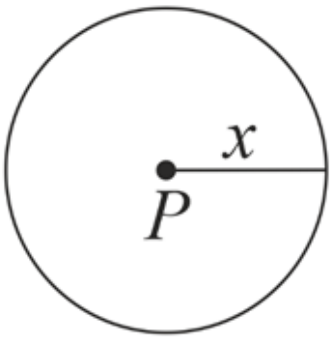
$I_{\mathrm{EF}}=\frac{I_{\mathrm{z}}}{2}=\frac{Ma^2}{6}$

$\therefore\ I_{\mathrm{AC}}=I_{\mathrm{EF}}.$

Q2

In telescope, the aperture diameter of objective lens is taken more so that more light can enter, so that image becomes sharper. Also due to more aperture diameter, the resolution power of telescope increases.

Q3



For a point source of power $=P$, then intensity at a point at a separation x from the source is

$I=\frac{\text{Power}}{\text{Area}}=\frac{P}{4\pi x^2}$

$\because$ Average intensity of e.m wave is given by,

$I = \frac{1}{2} C \in_0 E_o^2$

$\Rightarrow \frac{1}{2} C \in_0 E_o^2 = \frac{P}{4\pi x^2}$

$\Rightarrow E_o^2 = \frac{2P}{4\pi \in_0 C\, x^2}$

$\Rightarrow E_0 = \sqrt{\frac{2P}{4\pi \in_0 C\, x^2}}$

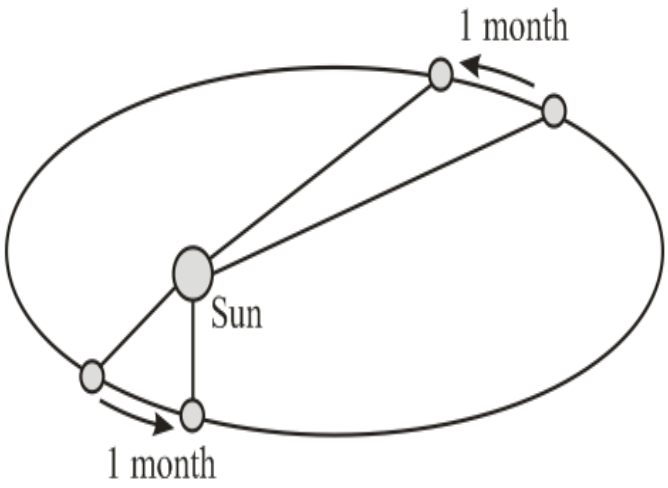
$\because \frac{1}{4\pi \in_0 \in} = 9 \times 10^9,\ P = 0.1\ W,\ x = 1\ m$

$C = \text{Speed of light} = 3 \times 10^8\ \text{m s}^{-1}$

$\Rightarrow E_0 = \sqrt{\frac{2 \times 0.1 \times 9 \times 10^9}{3 \times 10^8 \times 1^2}} = \sqrt{6} = 2.45\ \text{v m}^{-1}$

Q4

The kinetic energy is maximum when the particle is near to the sun because we know that from Kepler's law of areal speed, that the particle swept the equal area in equal time so at point A to cover equal-area it should move faster than other points.



Q5

Let,

$E =$energy of electron

$m =$mass of electron

$r =$ The radius of the orbit

then

$E \propto m$

$\Rightarrow \frac{E}{-13.6} = \frac{2m}{m}$

$\Rightarrow E = -27.2 \text{ eV}$

Also,

$r \propto \frac{1}{m}$

$\frac{r_0}{a_0} = \frac{m}{2m}$

$r_0 = \frac{a_0}{2}$

Q6

Here, the dimension of $\frac{a}{V^2}$ will be equal to pressure so $\frac{a}{(L^3)^2} = ML^{-1}T^{-2}$ [Principle of homogeneity]

$\therefore \quad [a] = \left[ML^5T^{-2}\right]$

Aliter:

According to gas equation, for one mol of a real gas.

$\left[P + \frac{a}{V^2}\right](V - b) = RT$

$PV + \frac{a}{V} - Pb + \frac{ab}{V^2} = RT$

As this equation is dimensionally correct, each term on either side will have same dimensions, i.e.,

$\left[\frac{a}{V}\right] = [PV]$

or $\quad [a] = \left[ML^{-1}T^{-2}\right]\left[L^3\right]\left[L^3\right] = \left[ML^5T^{-2}\right]$

and $[P \times\ b] = (PV)$

or $\quad [b] = [V] = [L]^3$

Note: Actually vander Waals equations for μ mol is

$\left[P + \frac{\mu^2 a}{V^2}\right][V - \mu b] = \mu RT$

So that $[\mu b] = [V] \quad i.\ e.,\ \ [b] = \left[L^3\mu^{-1}\right]$ with units m^3/mol

and $\left[\mu^2 a\right] = \left[PV^2\right] \quad i.\ e.,\ \ [a] = \left[ML^5T^{-2}\right]$ with units $J\ m^3/\text{mol}^2$

Q7

The expression of path difference is given as $\Delta x = d\sin\theta$, here θ is the angle relative to the incident direction.

For bright or maxima fringe, path difference is stated as$\Delta x = m\lambda$, here $m = 1, 2, 3, ..$

Giver here $D = 1\ \text{m},\ \text{d} = 1\ \text{mm}, \lambda = 5000\ \text{nm}$ and $m = 100$

For 100th maximum, $d\sin\theta = 100\lambda$

$\sin\theta = \frac{100\times5000\times10^{-9}}{1\times10^{-3}} = \frac{5\times10^{-4}}{10^{-3}} \times 0.5 = \frac{1}{2}$

$\theta = \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

We know that separation of 100th maxima from central maxima is given as $y = D\tan\theta$

Thus, distance $y = 1 \times \tan 30 = \frac{1}{\sqrt{3}}\ \text{m}.$

Q8

Volume remains constant.

So, the volume of big drop = volume of 1000 small drops

$\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^3$

$R = 10r$

Recall the formula of surface energy,

$E = T \times A$

$\therefore \ \frac{\text{Surface energy of big drop}}{\text{Surface energy of 1000 small drops}}$

$= \frac{4\pi R^2 T}{1000 \times 4\pi r^2 T}$

$= \frac{10 \times 10r}{1000r} \ \ [\because R = 10r]$

$= \frac{1}{10}$

Q9

The acceleration of the system $= \left(\frac{m_1 - m_2}{m_1 + m_2}\right) g$ and hence the tension T in the string is $\left(\frac{2m_1 m_2}{m_1 + m_2}\right) g$

The reading of the spring balance is T (in units of force) and T/g(in units of mass).

Q10

The relation between Y, η and B is $\frac{9}{Y} = \frac{1}{B} + \frac{3}{\eta}$

Where Y is Young's modulus, B is Bulk modulus and η is modulus of rigidity.

Q11

Focal length of combination

$\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2} \quad \ldots(\mathrm{i})$

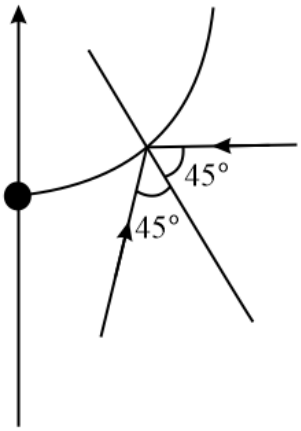
Using lens maker's formula

$\frac{1}{f_2}=(\mu_2-1)\left(\frac{1}{-R}-\frac{1}{\infty}\right)=-\frac{(\mu_2-1)}{R}$

$\frac{1}{f_1}=(\mu_1-1)\left(\frac{1}{\infty}-\frac{1}{-R}\right)=\frac{\mu_1-1}{R}$

or $\frac{1}{f_1}=\frac{(\mu_1-1)}{R}$

$\frac{1}{f_2}=-\frac{(\mu_2-1)}{R}$



Putting these values in Equation (i), we get

$\Rightarrow \frac{1}{f}=\frac{(\mu_1-1)}{R}-\frac{(\mu_2-1)}{R}$

$\Rightarrow \frac{1}{f}=\frac{(\mu_1-1-\mu_2+1)}{R}$

$\Rightarrow \frac{1}{f}=\frac{\mu_1-\mu_2}{R}$

$f=\frac{R}{(\mu_1-\mu_2)}$

Q12

When the ball of mass m falls from a height$h$, it reaches the surface of earth in time

$\sqrt{\frac{2h}{g}}$ and Its velocity is$v=\sqrt{2gh}$. It then moves in to the tunnel and reaches on the other side of earth and goes again upto a height h from that side of earth. The ball again returns back and thus executes periodic motion.

Outside the earth ball crosses distance h four times.

When the ball is in the tunnel at distance x from the centre of the earth, then gravitational force acting on ball is

$F=\frac{Gm}{x^2}\times\left(\frac{4}{3}\pi x^2\rho\right)=G\times\left(\frac{4}{3}\pi\rho\right)mx$

Mass of the earth, $M=\frac{4}{3}\pi R^2\rho$

or $\frac{4}{3}\pi\rho=\frac{\mathrm{M}}{\mathrm{R}^3}$

$\therefore \quad F=\frac{GMmx}{R^3}\ \ ie,\ F\propto x$

As this F is directed towards the centre of the earth i.e., the mean position so the ball will execute periodic motion about the centre of the earth

Here inertia factor=mass of ball$=m$

Spring factor$=\frac{GMm}{R^3}=\frac{gm}{R}$

$\therefore$ time period of oscillation of ball in the tunnel is

$T^{'}=2\pi\sqrt{\frac{\text{inertia factor}}{\text{spring factor}}}$

$=2\pi\sqrt{\frac{\mathrm{m}}{\mathrm{gm/R}}}=2\pi\sqrt{\frac{\mathrm{R}}{\mathrm{g}}}$

Time spent by ball outside the tunnel on both the sides will be

$4\sqrt{\frac{2h}{g}}$

Therefore, total time period of oscillation of ball is

$= 2\pi\sqrt{\frac{R}{g}} + 4\sqrt{\frac{2h}{g}}$

Q13

The rate of flow of charge gives electric current, while charge per unit change in potential gives capacitance. So,

$i = \frac{\mathrm{d}q}{\mathrm{d}t}$ & $C = \frac{\mathrm{d}q}{\mathrm{d}V}$

Thus, we get,

$i = \frac{q}{t} = \frac{CV}{t}$

$\Rightarrow t = \frac{CV}{i}$

$= \frac{20\times10^{-6}\times300}{10\times10^{-3}}$

$= 0.6\ \mathrm{s}$

Q14

Potential gradient of the potentiometer is given as $0.2\ \mathrm{V\ m^{-1}}$.

So, $x = 0.2\ \mathrm{V\ m^{-1}}$.

Here, the potential difference across the coil is, $V = IR = 0.1 \times 2 = 0.2\ \mathrm{V}$.

If the balancing length is l, then,

$V = xl$

$\Rightarrow 0.2 = 0.2 \times l$

$\Rightarrow l = 1\ \mathrm{m}$.

Q15

Rate of heat transfer in a slab is given as $Q = KA\frac{dt}{dx}$, where A is the area of the surface and $\frac{dt}{dx}$ is the temperature gradient.

Equating heat current in both slabs

$\frac{\mathrm{K}(\theta-\theta_1)}{3\mathrm{d}}=\frac{3\mathrm{K}(\theta_2-\theta)}{\mathrm{d}}$

$\theta-\theta_1=9\theta_2-9\theta$

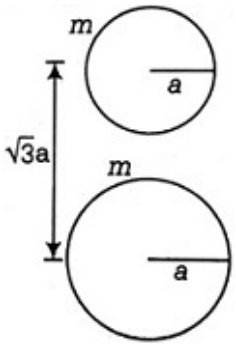
$10\theta=9\theta_2+\theta_1$

$\theta=\frac{9\theta_2+\theta_1}{10}$

Q16

Now, the gravitational field due to the ring at a distance $\mathrm{d}=\sqrt{3}\,\mathrm{a}$ on its axis is

$$\mathrm{E}=\frac{\mathrm{Gmd}}{\left(\mathrm{a}^2+\mathrm{d}^2\right)^{\frac{3}{2}}}=\frac{\sqrt{3}\mathrm{Gm}}{8\mathrm{a}^2}$$



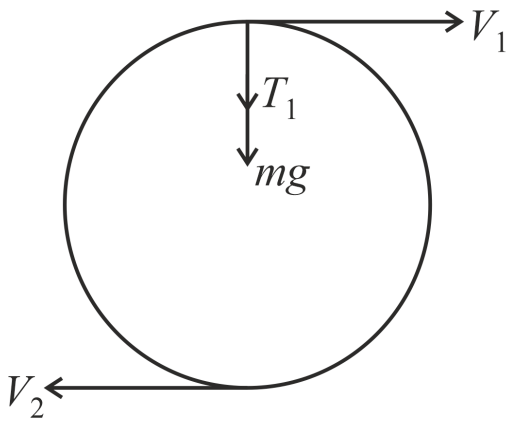
The force in a particle of mass M placed here is

$$\mathrm{F}=\mathrm{ME}=\frac{\sqrt{3}\mathrm{GMm}}{8\mathrm{a}^2}$$

This is also the force due to the sphere in the ring.

Q17

To just complete the loop, speed v_1 at the highest point must be minimum.



Applying Newton's 2nd law at the highest point.

$mg + T_1 = \frac{mv_1^2}{R}$.

For minimum v_1, tension must be minimum which can be zero, i.e. $T_1 = 0$.

$\therefore\ v_1 = \sqrt{Rg}$ ....(1).

Now from conservation of mechanical energy,

Loss in gravitational PE =Gain in KE

$\Rightarrow mg(2R) = \frac{1}{2}\mathrm{mv}_2^2 - \frac{1}{2}\mathrm{mv}_1^2$

$\Rightarrow v_2^2 = v_1^2 + 4gR$

$\Rightarrow v_2 = \sqrt{5gR}$ [Using (1)].

Hence, assertion is true, reason is false.

Q18

By the law of conservation of momentum $m\,u = (M + m)\nu$

$0.50 \times 2.00 = \Big(1 + 0\,.\,50\Big)\nu, \frac{1\,.\,00}{1.50} = \nu$

Initial K.E.$= \big(1/2\big) \times 0.50 \times \big(2\,.\,00\big)^2 = 1\,.\,00\mathrm{J}$

$$\text{Final K.E.} = \frac{1}{2} \times 1.50 \times \frac{1.00^2}{\left(1.50\right)^2} = \frac{1.00}{3.00} = 0.33$$

$$\therefore \text{Loss of energy} = 1.00 - 0.33 = 0.67 \text{ J}.$$

Q19

The given equation of alternating voltage is,

$$e = 200\sqrt{2}\sin 100t \ \ldots \left(\text{i}\right)$$

The standard equation of alternating voltage is,

$$e = e_0 \sin \omega t \ \ldots (\text{ii})$$

Comparing equations (i) and (ii), we get,

$$e_0 = 200\sqrt{2}\text{ V}, \omega = 100 \text{ rad s}^{-1}$$

The capacitive reactance is,

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times 1 \times 10^{-6}}\ \Omega$$

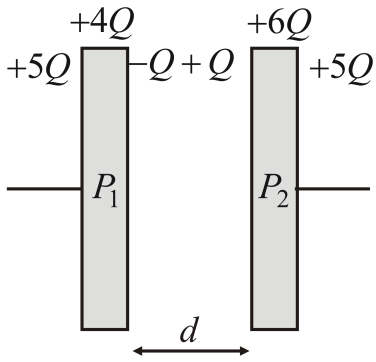
The rms value of current in the circuit is,

$$i_{rms} = \frac{v_{\text{rms}}}{X_C} = \frac{e_0/\sqrt{2}}{1/\omega C} = \frac{\left(200\sqrt{2}/\sqrt{2}\right)}{1/100 \times 10^{-6}}$$

$$= 200 \times 100 \times 10^{-6} \text{ A} = 2 \times 10^{-2} \text{ A} = 20 \text{ mA}$$

Q20

From the above questions charges on different plates will be,



When we connected both the plates with the wires, the charges will start flowing from one plate to the other and the final charges are as shown in the figure. Energy stored in the capacitor will be zero finally, i.e., $U_f = 0$, but initially, the energy will be stored in the electric field between the charges which can be given by,

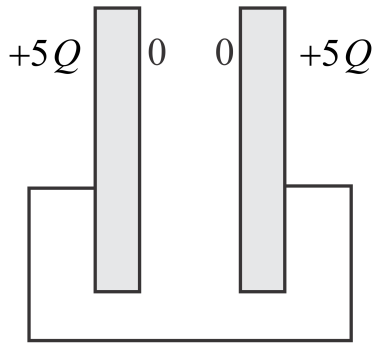
$$E = \frac{Q}{2A\varepsilon_0} + \frac{Q}{2A\varepsilon_0} = \frac{Q}{A\varepsilon_0}$$

So, the initial energy in the plates will be,

$$U_{\mathrm{i}} = \frac{1}{2}\varepsilon_0 E^2 \times Ad = \frac{1}{2} \times \varepsilon_0 \times \left(\frac{Q}{A\varepsilon_0}\right)^2 \times Ad$$

The energy lost in heat will be,

$$\Delta H = U_{\mathrm{i}} - U_{\mathrm{f}} = \frac{Q^2}{2A\varepsilon_0} d$$



Q21

Change in velocity = area under $a - t$ graph.

$v_{\mathrm{f}}-v_{\mathrm{i}}=\frac{1}{2}(4)\,(8)=16\ \mathrm{m\ s^{-1}}$.

$\therefore v_{\mathrm{f}}=v_{\mathrm{i}}+16=(5+16)\ \mathrm{m\ s^{-1}}=21\ \mathrm{m\ s^{-1}}$.

Q22

In series LCR, current is maximum at resonance.

$\therefore$ Resonant frequency, $\omega=\frac{1}{\sqrt{LC}}$

$\therefore\ \omega^2=\frac{1}{LC}$ or, $L=\frac{1}{\omega^2 C}$

Given, $\omega=1000\ \mathrm{s^{-1}}$ and $C=10\ \mu\mathrm{F}$

$\therefore\ L=\frac{1}{1000\times1000\times10\times10^{-6}}=0.1\ \mathrm{H}=100\ \mathrm{mH}\Rightarrow n=5$

Q23

The density can be computed by

$\rho=\frac{\mathrm{m}}{\mathrm{v}}=\frac{\mathrm{m}}{\frac{\pi \mathrm{d}^3}{6}}$

or $\rho\ \propto\ \frac{\mathrm{m}}{\mathrm{d}^3}$

$\therefore \frac{\Delta\rho}{\rho}\times100\%=\frac{\Delta \mathrm{m}}{\mathrm{m}}\times100\%\ +\ 3\frac{\Delta \mathrm{d}}{\mathrm{d}}\times100\%$

or $\frac{\Delta\rho}{\rho}\times100=2+(3\times3)=11$

Hence, the percentage error in measurement of density is 11%

Q24

$i=\frac{(12-8)}{(200+200)}\mathrm{A}=\frac{4}{400}=10^{-2}\ \mathrm{A}$

Power loss in each diode$=(4)\left(10^{-2}\right)\ \mathrm{W}=40\ \mathrm{mW}$

Q25

Given, initial temperature T1 = 80oC

Final temperature $T_2 = 50^oC$

Temperature of the surroundings $T_0 = 20^oC$

$$t_1 = 5 \text{ min}$$

According to Newton's law of cooling.

Rate of cooling, $\frac{dT}{dt} = k\left[\frac{T_1+T_2}{2} - T_0\right]$

$$\frac{(80-50)}{5} = k\left[\frac{80+50}{2} - 0\right]$$

$$\frac{30}{5} = k(65 - 20)$$

$$6 = k \text{ x } 45$$

or $k = \frac{6}{45} = \frac{2}{15}$

In second condition,

Initial temperature $T_1^{'} = 60^oC$

Final temperature $T_2^{'} = 30^oC$

$$t^{'} = ?$$

Now, $\frac{(60-30)}{t'} = \frac{2}{15}\left[\frac{60+30}{2} - 20\right]$

$\frac{30}{t'} = \frac{2}{15}(45 - 20)$

or $t' = \frac{30\times15}{2\times25}$

= 9 min.

Q26

By Mosely law $\frac{1}{\lambda} \propto (Z-1)^2$

$\frac{\lambda_2}{193} = \frac{(26-1)^2}{(29-1)^2}$

$\lambda_2 = \frac{193\times25^2}{28^2}$

λ_2 = 153.8 pm $\simeq$ 154 pm

Q27

Given,

the fundamental frequency of an open pipe, $L = 50$ cm is f,

the speed, $v = 350$ m s^{-1}

Now the fundamental frequency,

$f = \frac{v}{\lambda} = \frac{v}{2l} = \frac{350}{\left(2\times\frac{50}{100}\right)}$

$= 350$ Hz

Q28

Pressure of a gas is given by $P = \frac{1}{3}\frac{mN}{V}(v_{\text{rms}})^2$.

Where, m =mass of the gas,

$N =$ Number of gas molecules,

$V =$Volume of the vessel,

$v_{\text{rms}} =$ RMS speed of gas molecules.

So, $P_0 = \frac{1}{3}\frac{mN}{V}(v_{\text{rms}})^2$.

If the mass of all the molecules are halved and their speed is doubled,

$$P = \frac{1}{3}\frac{(m/2)N}{V}(2v_{\text{rms}})^2$$
$$\Rightarrow\ P = 2\left[\frac{1}{3}\frac{mN}{V}(v_{\text{rms}})^2\right]$$
$$\Rightarrow\ P = 2P_0$$

Therefore, $n = 2$.

Q29

Coverage distance d is

$d = \sqrt{2Rh}$

coverage area can be calculated as

$$\mathrm{A} = \pi \mathrm{d}^2 = 2\pi \mathrm{Rh}$$
$$\mathrm{A} = \pi \mathrm{d}^2 = 2 \times \pi \times 6400 \times 1000 \times 400$$
$$\mathrm{A} = \pi \mathrm{d}^2 = 1.6 \times 10^{10}$$

Q30

Given values,

the electric field, $\overrightarrow{E} = \left(y\hat{\mathrm{i}} + x\hat{\mathrm{j}}\right)\ \mathrm{N\ C^{-1}}$, charge, $q = 1\ \mathrm{C}$ and displacement,

$\overrightarrow{r}_{\mathrm{A}} = \left(2\hat{\mathrm{i}} + 2\hat{\mathrm{j}}\right)\ \mathrm{m}$ & $\overrightarrow{r}_{\mathrm{B}} = \left(4\hat{\mathrm{i}} + \hat{\mathrm{j}}\right)\ \mathrm{m}$

$d\overrightarrow{r} = \left(dx\hat{\mathrm{i}} + dy\hat{\mathrm{j}}\right)$

Use the relation between force and work done,

$dW = \overrightarrow{F} \cdot d\overrightarrow{r} = q\left(y\hat{\mathrm{i}} + x\hat{\mathrm{j}}\right) \cdot \left(dx\,\hat{\mathrm{i}} + dy\hat{\mathrm{j}}\right)$

$= q(dx \cdot y + dy \cdot x) = qd(xy)$

$\Rightarrow W = q\int_{2,\,2}^{4,\,1} d(xy) = 1 \times [xy]_{2,\,2}^{4,\,1} = 4 \times 1 - 2 \times 2 = 0$

Q31

Due to inert pair effect Pb has four electrons in its valence shell but it shows $+2$ oxidation state. In other words due to inert pair effect $+2$ oxidation state is more stable than $+4$ of Pb.

So, both Assertion and Reason are true and Reason is the correct explanation of Assertion.

Q32

Enzymes are a group of biomolecules which act as bio-catalyst, speeding up the reaction. These are made up of proteins. When such proteins encounter high temperature, it causes certain bonds to break in the structure, resulting in conformational change (denaturation). This denaturation makes the enzyme inactive and useless as catalyst. For an enzyme to be active, it may need support of non-protein constituents called cofactors, which bind to it, thus, making it catalytically active. These can be organic compounds, vitamins or mineral ions. The working of an enzyme binding to a single substrate was explained using a lock and key analogy given by Emil Fischer. In this analogy, the lock is the enzyme and the key is the substrate; on binding of substrate, the enzyme unlocks and changes the substrate to product. Also, this model explains the specificity of an enzyme as there is only one type of key to open the lock.

Q33

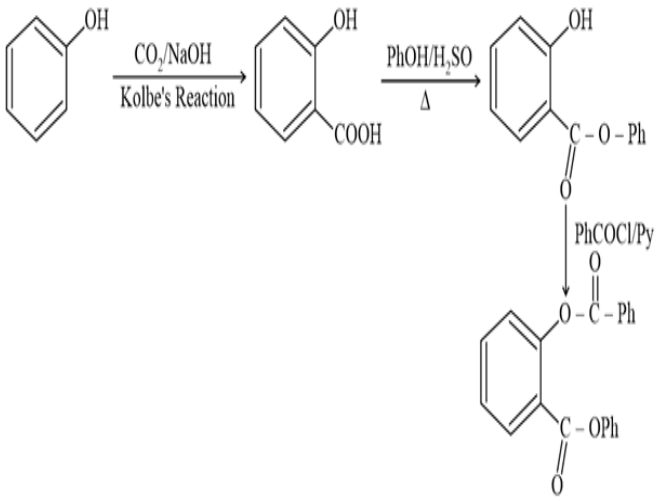
Statement-I is incorrect

$\mathrm{Be(OH)_2}$ dissolve in alkali due to it's amphoteric nature.

Statement-II is correct

Solubility of alkaline earth metal hydroxide in water increases down the group due to rapid decreases in lattice energy as compared to hydration energy.

Q34



In this reaction, first Kolbe's reaction product is formed and then phenol is attached to carboxylic acid in the presence of sulphuric acid to form an ester. This reaction is called esterification and again, an ester is formed when acyl chloride reacts with another alcohol.

Q35

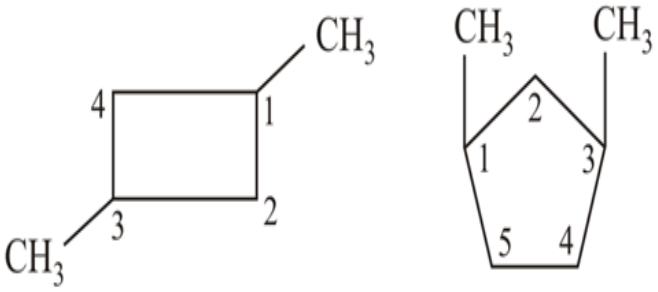
Water will pass through the membrane from the lower concentration region to the higher concentration region. Therefore, it will pass into the egg immersed in water, and out of the egg immersed in syrup.

Q36

Geometrical isomers (also known as cis-trans isomers or E-Z isomers), are chemical species with the same type and quantity of atoms as another species, but having a different geometrical structure. Atoms or groups exhibit different spatial arrangements on either side of a chemical bond or a ring structure.

In case of a closed chain, the ring must be substituted at least at two different positions.

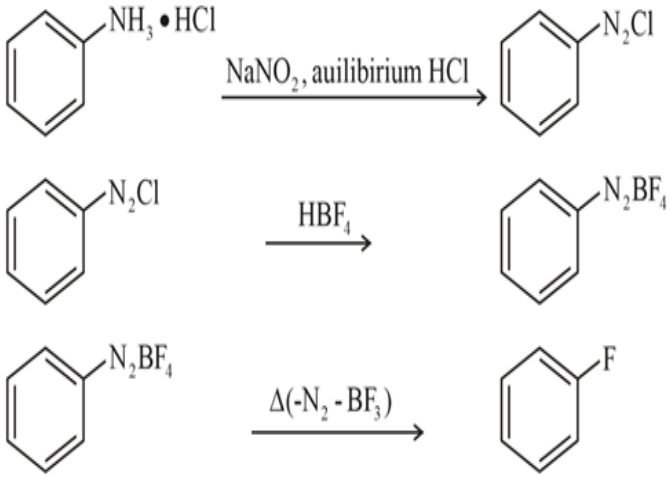
In options (a) and (b), there are no two ring carbons, containing different substituents, but in options (c) and (d), C-1 and C-3 have different substituents.



Q37

Aniline reacts with sodium nitrite and hydrochloric acid which forms benzene diazonium ion chloride which further reacts with hydrofluoric acid which gives fluoro benzene.

A is diazonium salt and B is flourobenzene.



Q38

In this particular case Zn is undergoing oxidation and Ag is undergoing reduction.

According to that first we will write the equation

For the cell
$Zn \left| Zn^{2+} \right. \| Ag^{+} \left| \right. Ag$

$E^{\circ}_{cell} = E^{\circ}_{cathode} + E^{\circ}_{anode}$

$1.56 = 0.8 + E^{\circ}_{Zn/Zn^{2+}}$

$E^{\circ}_{Zn/Zn^{2+}} = 0.76\ V$

Q39

Antacids: Antacid is a substance, which neutralizes stomach acidity and is used to relieve heartburn, indigestion or upset stomach.

Until 1970, the only treatment for acidity was the administration of antacids such as sodium hydrogencarbonate or a mixture of aluminium and magnesium hydroxide. But, excessive hydrogencarbonate can make the stomach alkaline and triggers the production of even more acid. Metal hydroxides are better alternatives because of being insoluble. These do not increase the pH above neutrality. These treatments control only symptoms, not the cause.

Antipyretics: These are the substances that reduce fever. Antipyretics cause the hypothalamus to override a prostaglandin-induced increase in temperature. The body then works to lower the temperature, which results in a reduction in fever.

Analgesics: An analgesic or painkiller is any member of the group of drugs used to achieve analgesia and relief from pain.

Antibiotics: Antibiotics also called antibacterials that are type of antimicrobial drugs used in the treatment and prevention of bacterial infections. They may either kill or inhibit the growth of bacteria.

Q40

The Wacker process is originally referred to the oxidation of ethylene to acetaldehyde by oxygen in water, where the catalyst is tetrachloropalladate(II).

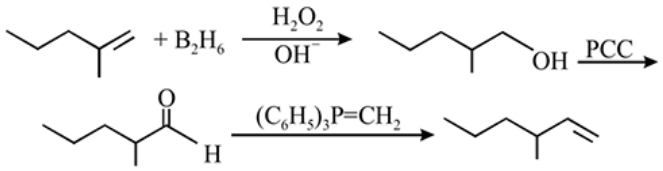
In the contact process, Platinum used to be the catalyst for this reaction, however, as it is susceptible to react with arsenic impurities in the sulphur feedstock, the preferred catalyst today is vanadium (V) oxide (V_2O_5).

In the Deacon's process, the reaction takes place at about 400 to $450^{\circ}\mathrm{C}$ in the presence of a variety of catalysts, which include copper chloride $(\mathrm{CuCl_2})$.

In Ziegler-Natta polymerisation, homogenous catalysts, usually based on complexes of Ti, Zr or Hf, are used. They are usually used in combination with different catalysts.

Q41

Here the final product is 3-methyl-1-hexene and it is formed as follows



Q42

In A, Iodine is oxidized from -1 to zero whereas in B, chlorine gets reduced from $+1$ to -1. So hydrogen peroxide acts as an oxidising agent in (A) and reducing agent in (B).

Q43

Mercury poisoning often produces a crippling and fatal disease called Minamata disease.

Q44

Statement 1: $\mathrm{B.P.}$ of chloroform $= 334\ \mathrm{K}$

$\mathrm{B.P.}$ of aniline $= 457\ \mathrm{K}$

thus can be seprated of simple distillation.

Statement 2 : Mixture of aniline and water seprated by simple distillation.

Q45

For first reaction:

$$\mathrm{E_1} = \frac{2.303\mathrm{RT_1\,T_2}}{(\mathrm{T_1}-\mathrm{T_2})}\log\frac{\mathrm{k_1^{'}}}{\mathrm{k_1}} \quad .....(\mathrm{i})$$

For second reaction:

$$\mathrm{E_2} = \frac{2.303\mathrm{RT_1\,T_2}}{(\mathrm{T_1}-\mathrm{T_2})}\log\frac{\mathrm{k_2^{'}}}{\mathrm{k_2}} \quad .....(\mathrm{ii})$$

Given: $\mathrm{E_1} > \mathrm{E_2}$

$$\Rightarrow \frac{2.303\mathrm{RT_1\,T_2}}{(\mathrm{T_1}-\mathrm{T_2})}\log\frac{\mathrm{k_1^{'}}}{\mathrm{k_1}} > \frac{2.303\mathrm{RT_1\,T_2}}{(\mathrm{T_1}-\mathrm{T_2})}\log\frac{\mathrm{k_2^{'}}}{\mathrm{k_2}}$$

$$\therefore \frac{\mathrm{k_1^{'}}}{\mathrm{k_1}} > \frac{\mathrm{k_2^{'}}}{\mathrm{k_2}}$$

Q46

Smelting : It is a form of extractive metallurgy; it's main use is to produce a metal from it's ore. Smelting uses heat and a chemical reducing agent like C, CO, etc. to decompose the ore driving off other elements as gases or slag and leaving just the metal behind.

Q47

Oxidation state of P in $\mathrm{H_4P_2O_7},\ \mathrm{H_4P_2O_5}$ and $\mathrm{H_4P_2O_6}$ is $5,\ 3$ & 4 respectively

$\mathrm{H_4P_2O_7}$

$2\mathrm{x} + 4(+1) + 7(-2) = 0$

$\mathrm{x} = +5$

$\mathrm{H_4\underline{P_2}O_5}$

$2\mathrm{x} + 4(+1) + 5(-2) = 0$

$\mathrm{x} = +3$

$\mathrm{H_4\underline{P_2}O_6}$

$2x + 4(+1) + 6(-2) = 0$

$x = +4$

Q48

When O_2 changes to O_2 : $\left(\sigma 1s\right)^2(\sigma^* 1s)^2\left(\sigma 2s\right)^2(\sigma^* 2s)^2(\sigma 2p_z)^2\left(\pi 2p_x^2 = \pi 2p_y^2\right)$

$\left(\pi^* 2p_x^1 = \pi^* 2p^1 y\right)$

Bond order : $\frac{1}{2}[N_b - N_a] = \frac{1}{2}\left[10 - 6\right] = 2$

Magnetic moment $\left(\mu\right)$: $\sqrt{n(n+2)}BM = \sqrt{2(2+2)}$

$= 2.828\,\text{BM}$

O_2^+ : $\left(\sigma 1s\right)^2(\sigma^* 1s)^2\left(\sigma 2s\right)^2(\sigma^* 2s)^2(\sigma 2p_z)^2\left(\pi 2p_x^2 = \pi 2p^2 y\right)$

$\left(\pi^* 2p_x^1 = \pi^* 2p_y^0\right)$

Bond order : $\frac{1}{2}(N_b - N_a) = \frac{1}{2}\left(10 - 5\right) = 2.5$

Magnetic moment $\left(\mu\right)$: $\sqrt{n(n+2)}BM$

$= \sqrt{1(1+2)} = 1.732\,\text{BM}$

Q49

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O;\ \ \Delta H = -188$ kJ/mol ...(i)

$H_2 + O_2 \rightarrow H_2O_2;\ \ \Delta H = -286$ kJ/mol ...(ii)

By Eq.(i) $-$ (ii)

$2H_2O_2 \rightarrow 2H_2O + O_2;$

$\Delta H = 2(-188) - 2(-286) = +196$ kJ

Q50

Nylon-66 is made from adipic acid.

Nylon 66 is synthesized by polycondensation of hexamethylenediamine and adipic acid. Equivalent amounts of hexamethylenediamine and adipic acid are combined with water in a reactor.

Q51

V = 1 L moles = Molar

$$\begin{array}{ccccc} 2\,\mathrm{H_2\,S(g)} & \rightleftharpoons & 2\mathrm{H_2(g)} & + & \mathrm{S_2(g)} \\ 1-2\mathrm{x} & & 2\mathrm{x} & & \mathrm{x} \end{array}$$

Total moles $= 1 - 2\mathrm{x} + 2\mathrm{x} + \mathrm{x} = 1 + \mathrm{x}$

$\mathrm{K_c} = 4 \times 10^{-6} = \frac{(2\mathrm{x})^2(\mathrm{x})}{(1-2\mathrm{x})^2} = \frac{4\mathrm{x}^3}{1}$

$(1 - 2\mathrm{x} \approx 1)$

$\mathrm{x}^3 = 10^{-6}$

$\mathrm{x} = 10^{-2}$

$2\mathrm{x} = 2 \times 10^{-2}$

% Dissociation = 2%

Q52

In Hoffmann's mustard oil reaction, the primary amine reacts with alcoholic carbon disulphide and then heated with excess of mercuric chloride and formed the isothiocyanate. Primary amines can only give Hoffmann's mustard oil reaction. Primary amine is the amine in which nitrogen atom is connected only with one carbon atom and two hydrogen atoms.

Q53

$3\ \mathrm{Pb(NO_3)_2} + \mathrm{Cr_2(SO_4)_3} \rightarrow 3\,\mathrm{PbSO_4} + 2\,\mathrm{Cr(NO_3)_3}$

$35\ \mathrm{ml}$	$20\ \mathrm{ml}$
$0.15\ \mathrm{M}$	$0.12\ \mathrm{M}$

$= 5.25\ \mathrm{m.mol} = 2.4\ \mathrm{m.mol} \quad 5.25\ \mathrm{m.mol}$

$= 5.25 \times 10^{-3}\ \mathrm{mol}$

Therefore moles of $\mathrm{PbSO_4}$ formed

$= 5.25 \times 10^{-3} = 525 \times 10^{-5}$

Q54

$\Delta \mathrm{T_f} = \mathrm{iK_f m}$

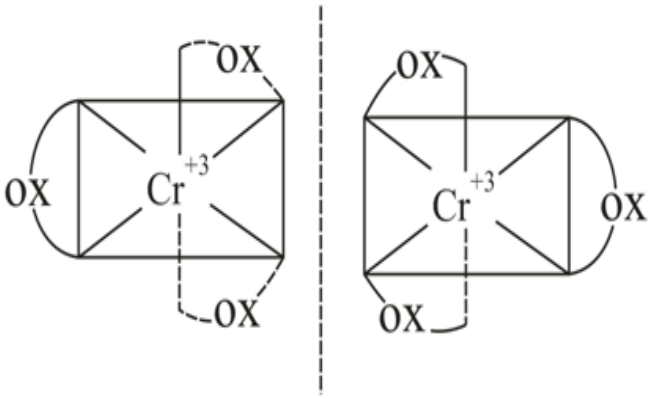
$0.0557 = \mathrm{i} \times 1.86 \times 0.01$

$\mathrm{i} = 3$, means complex will give 3-ions

Hence, formula should be $\left[\mathrm{Co(NH_3)_5\,Br}\right]\mathrm{Br_2}$

Q55

The number of optical isomers for $[\mathrm{Cr(C_2O_4)s}]^{3-}$ is two.



Q56

Solute in 200 g of 30% solution $=60$ g

Solute in 300 g of 20% solution $=60$ g

total grams of solute $=120$ g

total grams of solution $=200+300=500$ g

% of solute in the final solution $=\frac{\text{Wt. of solute}}{\text{Wt. of solution}} \times 100$

% of solute in the final solution $=\frac{120}{500} \times 100=24\%$

Q57

$\text{Energy}_{\text{photon}}\left(\text{eV}\right)=\frac{1240}{\text{wavelength(nm)}}$

$\text{Energy}_{\text{photon}}=\frac{1240}{300}=4.1\ \text{eV}$

According to photoelectric effect, those metals will show this effect which have work function value less than the energy of photon.

So on comparing energy of photon and work function values for the given metal we can conculde that in

$\text{Li},\ \text{Na},\ \text{K}\ \&\ \text{Mg}$ i.e. in 4 metals , photo emission will take place.

Q58

$\text{HA}+\text{OH}^{-} \rightleftharpoons \text{H}_2\text{O}+\text{A}_{\text{aq}}^{-1}\ ;\ \text{K}_{\text{eq}}=10^{9}$

Weak acid st. base

Then for hydrolysis reaction

$\mathrm{A}_{\mathrm{aq}}^{-}+\mathrm{H}_2\mathrm{O} \rightleftharpoons \mathrm{HA}+\mathrm{OH}_{\mathrm{aq}}^{-} ; \mathrm{K}_{\mathrm{h}}=10^{-9}$

$\left[\mathrm{OH}^{-}\right]=\sqrt{\mathrm{K}_{\mathrm{h}} \cdot \mathrm{C}}=\sqrt{10^{-9} 10^{-1}}=10^{-5}$

$\mathrm{pOH}=-\log 10^{-5}=5$

$\therefore \mathrm{pH}=14-\mathrm{pOH}=9$

Q59

$1 \text{ mole } \mathrm{KBr}\ (=119\,\mathrm{gm})$ have $\frac{10^{-5}}{100}$ moles SrBr_2 and hence, 10^{-7} moles cation vacancy (as $1\,\mathrm{Sr}^{2+}$ will result 1 cation vacancy)

$\therefore$ Required number of cation vacancies

$=\frac{10^{-7} \times 6.023 \times 10^{23}}{119}=5.06 \times 10^{14} \simeq 5 \times 10^{14}$

Q60

Half-life is 500 minutes for all concentrations of sugar. Half-life is independent of the concentration of the sugar. So reaction is 1st order with respect to sugar. For H^{+} concentration

$\frac{t_1}{t_2}=\left(\frac{a_2}{a_1}\right)^{n-1}$

$\frac{500}{50}=\left(\frac{10^{-6}}{10^{-5}}\right)^{n-1}$

Therefore, $n=0$

Q61

Let,

$t=f(x)=\log_a\left(x^2\right)=2\log_a|x|=2\log_a x \text{ as } x>0$

$\Rightarrow t = 2\log_a x$

$\Rightarrow \frac{t}{2} = \log_a(x)$

$\Rightarrow x = a^{\frac{t}{2}}$

$f^{-1}(x) = a^{\frac{x}{2}}$

$f^{-1}(b+c) = a^{\frac{b+c}{2}} = a^{\frac{b}{2}}a^{\frac{c}{2}} = f^{-1}(b)\cdot f^{-1}(c)$

Q62

$\vec{r} \times \vec{a} = \vec{b} \times \vec{r} \Rightarrow \vec{r} \times \left(\vec{a} + \vec{b}\right) = 0$

$\vec{r} = \vec{\lambda}\left(\vec{a} + \vec{b}\right) \Rightarrow \vec{r} = \vec{\lambda}\left(\hat{i} + 2\hat{j} - 3\hat{k} + 2\hat{i} - 3\hat{j} + 5\hat{k}\right)$

$\vec{\mathrm{r}} = \vec{\lambda}\left(3\hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\widehat{\mathrm{k}}\right) \ \ldots(1)$

$\vec{r} \cdot \left(\alpha\hat{i} + 2\hat{j} + \hat{k}\right) = 3$

Put $\vec{r}$ from (1) $\alpha\lambda = 1 \ \ldots(2)$

$\vec{r} \cdot \left(2\hat{i} + 5\hat{j} - \alpha\hat{k}\right) = -1$

Put $\vec{r}$ from (1) $2\lambda\alpha - \lambda = 1 \ \ldots(3)$

Solve (2) and (3)

$\alpha = 1, \lambda = 1$

$\Rightarrow \ \vec{\mathrm{r}} = 3\hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\widehat{\mathrm{k}}$

$\left|\vec{r}\right|^2 = 14 \ \& \ \alpha = 1$

$\alpha + \left|\vec{r}\right|^2 = 15$

Q63

Basically, we need to find the intersection of the ray and mirror.

Let us assume $A'(x_1,\ y_1)$ is the image of $A(2,\ 3)$ with respect to $x + y = 0$.

$\Rightarrow A^{'}(-3,\ -2)$

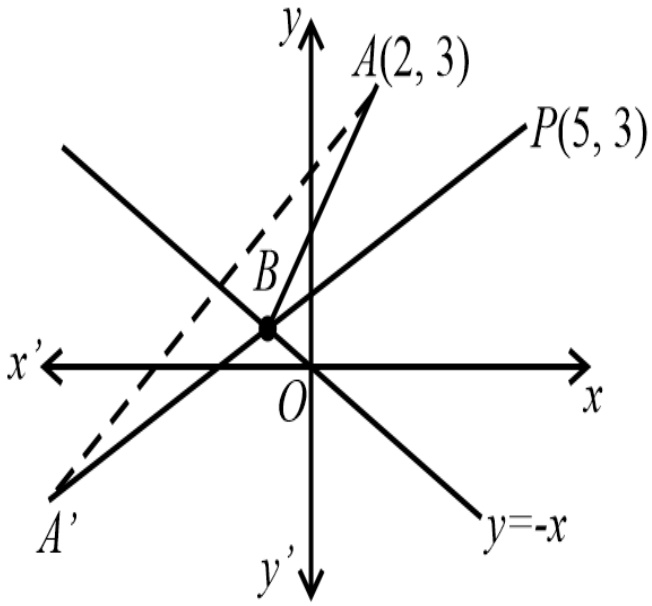
Now, we can say that B is the point of intersection of PA' with $x + y = 0$.

The slope of PA' is $\frac{5}{8}$.

Hence, the equation of the line PA' is given as,

$y - 3 = \frac{5}{8}(x - 5)$

$\Rightarrow 5x - 8y - 1 = 0$



Now, on solving the equations $y+x=0$ and $5x-8y-1=0$, we will get the point B as $\left(\frac{1}{13},\ -\frac{1}{13}\right)$

Q64

Given equation are $x+y+z=6,\ x+2y+3z=10$ and $x+2y+\lambda z=10$

Since, it is consistent.

$$\therefore \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow \quad 1(2\lambda-6)-1(\lambda-3)+1(2-2)=0$$

$$\Rightarrow \quad \lambda-3=0 \ \Rightarrow \ \lambda=3$$

Q65

$$\cot^{-1}\frac{5}{\sqrt{3}}+\cot^{-1}\frac{9}{\sqrt{3}}+\cot^{-1}\frac{15}{\sqrt{3}}+\cot^{-1}\frac{23}{\sqrt{3}}+\ldots\ldots\infty$$

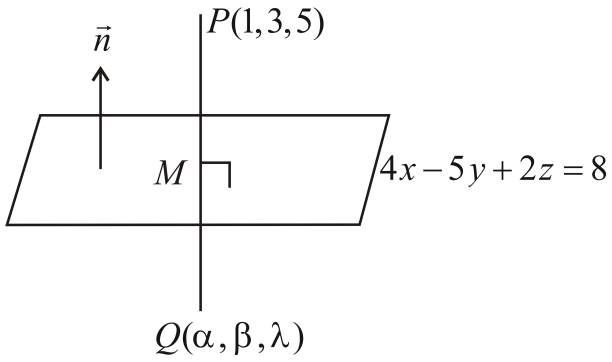
$$=\tan^{-1}\left(\frac{\frac{2}{\sqrt{3}}-\frac{1}{\sqrt{3}}}{1+\frac{2}{\sqrt{3}}\times\frac{1}{\sqrt{3}}}\right)+\tan^{-1}\left(\frac{\frac{3}{\sqrt{3}}-\frac{2}{\sqrt{3}}}{1+\frac{3}{\sqrt{3}}\times\frac{2}{\sqrt{3}}}\right)+\ldots\ldots+\ldots\ldots+\tan^{-1}\left(\frac{\frac{n+1}{\sqrt{3}}-\frac{n}{\sqrt{3}}}{1+\frac{n+1}{\sqrt{3}}\times\frac{n}{\sqrt{3}}}\right)+\ldots\ldots$$

$$=\left(\tan\frac{2}{\sqrt{3}}-\tan^{-1}\frac{1}{\sqrt{3}}\right)+\left(\tan^{-1}\frac{3}{\sqrt{3}}-\tan^{-1}\frac{2}{\sqrt{3}}\right)+\left(\tan^{-1}\frac{4}{\sqrt{3}}-\tan^{-1}\frac{3}{\sqrt{3}}\right)+\ldots\ldots+\left(\tan^{-1}\frac{n+1}{\sqrt{3}}-\tan^{-1}\frac{n}{\sqrt{3}}\right)$$

$$\Rightarrow \ S_n=\tan^{-1}\frac{n+1}{\sqrt{3}}-\tan^{-1}\frac{1}{\sqrt{3}}$$

$$S_\infty=\frac{\pi}{2}-\frac{\pi}{6}=\frac{\pi}{3}$$

Q66



Point Q is image of point P w.r.to plane, M is mid point of P and Q, lies in plane

$M\left(\frac{1+\alpha}{2}, \frac{3+\beta}{2}, \frac{5+\gamma}{2}\right)$

$4x - 5y + 2z = 8$

$4\left(\frac{1+\alpha}{2}\right)-5\left(\frac{3+\beta}{2}\right)+2\left(\frac{5+\gamma}{2}\right)= 8 \ \ldots(1)$

Also PQ perpendicualr to the plane

$\Rightarrow \overrightarrow{PQ} \| \overrightarrow{n}$

$\frac{\alpha-1}{4} = \frac{\beta-3}{-5} = \frac{\gamma-5}{2} = k$ (let)

$$\left.\begin{aligned}\alpha &= 1+4k\\ \beta &= 3-5k\\ \gamma &= 5+2k\end{aligned}\right\} \ldots(2)$$

use (2) in (1)

$2(1+4k)-5\left(\frac{6-5k}{2}\right)+(10+2k)= 8$

$k = \frac{2}{5}$

from (2) $\alpha = \frac{13}{5}, \beta = 1, \gamma = \frac{29}{5}$

$5(\alpha+\beta+\gamma)= 13+5+29 = 47$

Q67

Let, $M=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$\therefore$ Sum of the diagonal elements, $\operatorname{Tr}\left(M^T M\right)=a^2+b^2+c^2+d^2+e^2+f^2+g^2+h^2+i^2=5$, where entries are $\{0,1,2\}$

Only two cases are possible.

(I) Five entries are 1 and other fourare 0

$\therefore \quad {}^9C_5 \times 1$

(II) One entry is 2, one entry is 1 and others are 0.

$\therefore \quad {}^9C_2 \times 2$

Total cases$=126+72=198$.

Q68

The given statement is $p \rightarrow(q \wedge r)$

Where,

p : The weather is fine.

q : My friends will come.

r : We will go for a picnic.

The contrapositive is $(\backsim(q \wedge r)) \rightarrow \backsim p$

Now, using De Morgan's Law, we have

$(\backsim q \vee \backsim r) \rightarrow \backsim p$

If my friends do not come or we do not go for picnic then weather will not be fine.

Q69

Let the probability of getting any number other than 5 is P, then the probability of getting 5 is 5P.

$\because \mathrm{P}+\mathrm{P}+\mathrm{P}+\mathrm{P}+\mathrm{P}+5\mathrm{P}=1 \Rightarrow \mathrm{P}=\frac{1}{10}$

Expected income per throw $=\frac{5}{10} \times 5+\left(\frac{1}{10} \times 8\right) 5$

$=\frac{65}{10}=6.5$

Q70

Given Expression is, $\left(x+{}^{2n+1}C_0\right)\left(x+{}^{2n+1}C_1\right)\left(x+{}^{2n+1}C_2\right) \ldots . \left(x+{}^{2n+1}C_n\right)$

If P is coefficient of x^n then,

$P={}^{2n+1}C_0+{}^{2n+1}C_1+{}^{2n+1}C_2+\ldots . .+{}^{2n+1}C_n$ -------(1)

$\Rightarrow P={}^{2n+1}C_{2n+1}+{}^{2n+1}C_{2n}+{}^{2n+1}C_{2n-1}+\ldots . .+{}^{2n+1}C_{n+1}$ ------(2)$(\because {}^nC_r={}^n C_{n-r})$

adding (1) and (2)

$2P=\left({}^{2n+1}C_0+{}^{2n+1} C_1+\ldots . . .+{}^{2n+1} C_{2n+1}\right)$

$2P=2^{2n+1}$

$\therefore \quad P=2^{2n}$

Q71

(i) aRa, then GCD of a and a is a.

$\therefore$ R is not reflexive.

(ii) $aRb \Rightarrow bRa$

If GCD of a and b is 2, then GCD of b and a is 2.

$\therefore$ R is symmetric.

(iii) $aRa,\ bRc \Rightarrow cRa$

If GCD of a and b is 2 and GCD of b and c is 2, then it is need not to be GCD of c and a is 2.

$\therefore$ R is not transitive.

Q72

$1^2 + 2^2 - 3^2 + 4^2 + 5^2 - 6^2 + \ldots + 30$ terms

$= 1^2 + 2^2 + 3^2 + 4^2 + \ldots + 30^2 - 2\left[3^2 + 6^2 + 9^2 + \ldots + 30^2\right]$

$= 1^2 + 2^2 + 3^2 + 4^2 + \ldots + 30^2 - 2 \times 3^2\left[1^2 + 2^2 + 3^2 + \ldots + 10^2\right]$

$= \frac{30 \times 31 \times 61}{6} - \frac{18 \times 10 \times 11 \times 21}{6}$ $\left(\because \sum_{r=1}^{n} r^2 = \frac{n\,(n+1)\,(2n+1)}{6}\right)$

$= 5 \times 31 \times 61 - 3 \times 10 \times 11 \times 21$

$= 9455 - 6930$

$= 2525$

Q73

Let $z = x + iy,\ \bar{z} = x - iy$

$(2iy)^2 = 4(x^2 + y^2) - 12$

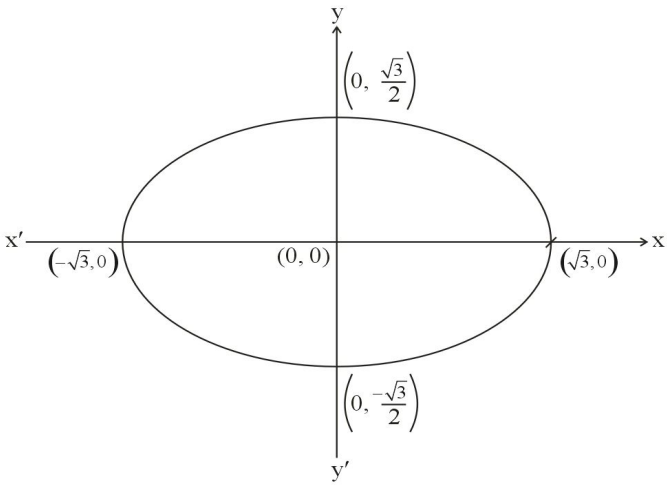
$-4y^2 = 4(x^2 + y^2) - 12$

$x^2 + 2y^2 = 3$

$\frac{x^2}{3} + \frac{y^2}{\frac{3}{2}} = 1$

It represents an ellipse

So,



$|z|_{max} = \sqrt{3}$

Q74

Equation of chord of contact AB is

$5x - 5y = 1$

$\Rightarrow x-y=1$

Therefore,

$x^2+(x-1)^2=5$

$\Rightarrow 2x^2-2x-4=0$

$\Rightarrow x=2,\ -1$

$\Rightarrow y=1,\ -2$

Hence, $A(2,1)$ and $B(-1,-2)$.

And

$P\equiv\left(\sqrt{5}\cos\theta,\sqrt{5}\sin\theta\right)$

Let the locus of orthocenter be (h,k). Now, we know that the centroid divides the line joining orthocenter and circumcenter in $2:1$. Therefore,

$h=2-1+\sqrt{5}\cos\theta=1+\sqrt{5}\cos\theta$

$k=1+(-2)+\sqrt{5}\sin\theta=-1+\sqrt{5}\sin\theta$

$\Rightarrow (h-1)^2+(k+1)^2=5$

Hence, the required locus is

$(x-1)^2+(y+1)^2=5$

Q75

$n=2^\alpha\cdot 3^\beta$

Given number of divisors$=12$

$\Rightarrow(\alpha+1)(\beta+1)=12$

For $2n = 2^{\alpha+1} \cdot 3^{\beta}$

Given number of divisors$= 15$

$\Rightarrow(\alpha+2)(\beta+1)= 15$

Dividing both$= \frac{\alpha+1}{\alpha+2} = \frac{4}{5}$

$\Rightarrow 5\alpha + 5 = 4\alpha + 8$

$\Rightarrow \alpha = 3$

So, $\beta = 2$.

Now for $3n = 2^{\alpha} \cdot 3^{\beta+1}$

Number of divisors$=(\alpha+1)(\beta+2)$

$= 4 \times 4 = 16$

Q76

Let, $t = 2^{11x}$

$\Rightarrow \frac{\left(2^{11x}\right)^3}{2^2} + 2^{11x}.2^2 = \left(2^{11x}\right)^2.2 + 1$

$\Rightarrow \frac{t^3}{4} + 4t = 2t^2 + 1$

$\Rightarrow t^3 - 8t^2 + 16t - 4 = 0$

Cubic in t has roots t_1, t_2, t_3

i.e. $t_1t_2t_3 = 4 \Rightarrow 2^{11x_1}.2^{11x_2}.2^{11x_3} = 4$

$\Rightarrow 2^{11(x_1+x_2+x_3)} = 2^2$

$\Rightarrow 11(x_1 + x_2 + x_3)= 2 \Rightarrow x_1 + x_2 + x_3 = \frac{2}{11}$

Q77

$f'(x)= 0$ must have two real and unequal roots

$f'(x)= 3x^2 - 6(2\lambda - 1)x + 6\lambda = 0$ must have

$D > 0 \Rightarrow 36(2\lambda - 1)^2 - 72\lambda > 0$

$\Rightarrow 4\lambda^2 + 1 - 4\lambda - 2\lambda > 0 \Rightarrow 4\lambda^2 - 6\lambda + 1 > 0,\ \lambda \in \left(-\infty,\ 3 - \sqrt{5}\right) \cup \left(3 + \sqrt{5},\ \infty\right)$

Q78

Given, $\sin\ 2x\left(\frac{dy}{dx} - \sqrt{\tan\ x}\right) - y = 0$

or, $\frac{dy}{dx} = \frac{y}{\sin\ 2x} + \sqrt{\tan\ x}$

or, $\frac{dy}{dx} - y\ \text{cosec}\ 2x = \sqrt{\tan\ x}$ ....(i)

Now, integrating factor (I.F) $(\text{I. F}) = e^{\int -\text{cosec}\ 2x}$

or, $\text{I. F} = e^{-\frac{1}{2}\log\ |\tan\ x|} = e^{\log\ \left(\sqrt{\tan\ x}\right)^{-1}}$

$= \frac{1}{\sqrt{\tan\ x}} = \sqrt{\cot\ x}$

Now, general solution of eq. (i) is written as

$y(\text{I. F.}) = \int Q(\text{I. F.})dx + c$

$\therefore\ y\sqrt{\cot\ x} = \int \sqrt{\tan\ x} \cdot \sqrt{\cot\ x} dx + c$

$\therefore\ y\sqrt{\cot\ x} = \int 1.\, dx + c$

$\therefore\ y\sqrt{\cot\ x} = x + c$

Q79

We know that

$\frac{1}{x} - 1 < \left[\frac{1}{x}\right] \leq \frac{1}{x}$

Hence,

$\frac{1+2+3...+15}{x} - 15 < \left[\frac{1}{x}\right] + \left[\frac{2}{x}\right] + ... + \left[\frac{15}{x}\right] \leq \frac{1+2+...+15}{x}$

$\Rightarrow \lim\limits_{x \to 0^+} x\left(\left[\frac{1}{x}\right] + \left[\frac{2}{x}\right] + ... + \left[\frac{15}{x}\right]\right) = 15\left(\frac{16}{2}\right)$

$= 120$

Q80

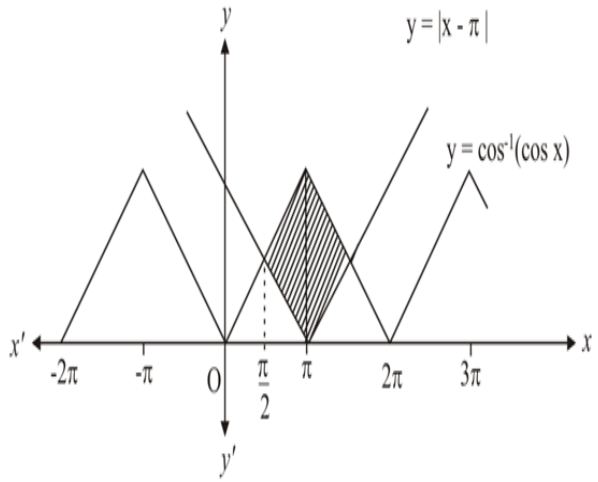
$T : y = mx \pm \sqrt{9\,m^2 + 1}\,\therefore 2 = \frac{\sqrt{9\,m^2+1}}{\sqrt{1+m^2}} \Rightarrow m = -\frac{\sqrt{3}}{\sqrt{5}}$
$e = \frac{2\sqrt{2}}{3}\,\therefore F\left(2\sqrt{2}, 0\right), RS : y = -\frac{\sqrt{3}}{\sqrt{5}}\left(x - 2\sqrt{2}\right) \Rightarrow \sqrt{5}y + \sqrt{3}x = 2\sqrt{6}$Let T be foot of perpendicular from $O(0,0)$ to RS Then,

$OT = \frac{2\sqrt{6}}{2\sqrt{2}} = \sqrt{3}\ \ \therefore\ R'S' = 2\sqrt{4-3}$

Q81

Given, $f(x) = \cos^{-1}(\cos x) = \begin{cases} x, & x \in [0,\ \pi] \\ 2\pi - x, & x \in [\pi,\ 2\pi] \end{cases}$

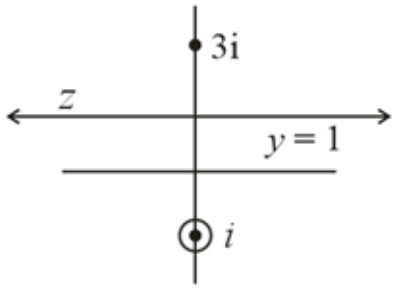
Graph of $y = f(x)$ has been shown in the figure.



$\therefore\ A = 2\int_{\frac{\pi}{2}}^{\pi}\ [x - (\pi - x)]dx = 2\ \left[x^2 - \pi x\right]_{\frac{\pi}{2}}^{\pi} = \frac{\pi^2}{2}$ sq. units

Hence, $n = 2$.

Q82



$\omega = z\bar{z} - 2z + 2$

$\left|\frac{z+i}{z-3i}\right| = 1$

$\Rightarrow |z+i| = |z-3i|$

$\Rightarrow z = x + i,\ x \in R$

$\omega = (x+i)(x-i) - 2(x+i) + 2$

$= x^2 + 1 - 2x - 2i + 2$

$\text{Re}(\omega) = x^2 - 2x + 3$

For min $(\text{Re}(\omega)), x = 1$

$\Rightarrow \omega = 2 - 2i = 2(1-i) = 2\sqrt{2}e^{-i\frac{\pi}{4}}$

$\omega^n = \left(2\sqrt{2}\right)^n e^{-i\frac{n\pi}{4}}$

For real and minimum value of n,

$n = 4$

Q83

$\frac{b}{c} = \frac{c}{b} = (\text{integer})$

$b^2 = ac \ \Rightarrow \ \ c = \frac{b^2}{a}$

$\frac{a+b+c}{3} = b + 2$

$a + b + c = 3b + 6 \Rightarrow a - 2b + c = 6$

$a - 2b + \frac{b^2}{a} = 6 \ \Rightarrow 1 - \frac{2b}{a} + \frac{b^2}{a^2} = \frac{6}{a}$

$\left(\frac{b}{a} - 1\right)^2 = \frac{6}{a} \ \Rightarrow \ \ a = 6$ only

Q84

Let 2 distinct number are a, b

$\frac{a+b}{2} = 60 \Rightarrow a + b = 120$

if $a = 0$ then $b = 120$

$a = 1$ then $b = 119$

.

.

.

$a = 59$ then $b = 61$

favourable case $= 60$

Total cases $= {}^{181}C_2$

Required probability $= \frac{60}{{}^{181}C_2} = \frac{2}{543} = k$

$\Rightarrow 1086k = 4$

Q85

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 4 \\ 0 & 1 & 2 \end{vmatrix} = \hat{i}(-8) - \hat{j}(6) + \hat{k}(3)$$

$$= -8\hat{i} - 6\hat{j} + 3\hat{k}$$

$$\left[\vec{a} \quad \vec{b} \quad \vec{c}\right] = \left(\vec{a} \times \vec{b}\right) \cdot \vec{c} \leq \left|\vec{a} \times \vec{b}\right| \left|\vec{c}\right|$$

$\Rightarrow$ maximum value of $\left[\vec{a} \quad \vec{b} \quad \vec{c}\right] = \left|\vec{a} \times \vec{b}\right| \left|\vec{c}\right|$

$$\Rightarrow k = \sqrt{64 + 36 + 9}.1 = \sqrt{109}$$

$$\Rightarrow k^2 - 50 = 109 - 50 = 59$$

Q86

Let $M = \left(P^{-1}AP - I\right)^2$

$$= \left(P^{-1}AP\right)^2 - 2P^{-1}AP + I$$

$$= P^{-1}A^2P - 2P^{-1}AP + I$$

$$PM = A^2P - 2AP + P$$

$$= \left(A^2 - 2\ A.I + I^2\right)P$$

$$\Rightarrow \text{Det}(PM) = \text{Det}\left((A - I)^2 \times P\right)$$

$$\Rightarrow \text{Det}(PM) == \text{Det}\,(A - I)^2 \times \text{Det}(P)$$

$$\Rightarrow \text{Det}\ M == (\text{Det}(A - I))^2$$

$$\text{Now } A-I=\begin{bmatrix}1 & 7 & w^2\\ -1 & -w-1 & 1\\ 0 & -w & -w\end{bmatrix}$$

$$\text{Det}(A-I)=\left(w^2+w+w\right)+7(-w)+w^3=-6w$$

$$\text{Det}\left((A-I)\right)^2=36w^2$$

$$\Rightarrow \alpha=36$$

Q87

We have,

$$n_1=20,\ \bar{x}_1=50,\ \sigma_1^2=1$$

$$n_1=40,\ \bar{x}_1=50,\ \sigma_2^2=4$$

Then,

$$\bar{x}=\frac{n_1\bar{x}_1+n_2\bar{x}_2}{n_1+n_2}$$

$$\Rightarrow \bar{x}=\frac{20\times50+40\times50}{20+40}=50$$

Then,

$$d_1=\bar{x}-\bar{x}_1=0$$

$$d_1=\bar{x}-\bar{x}_2=0$$

Then,

$$\sigma^2=\frac{n_1\left(\sigma_1^2+d_1^2\right)+n_2\left(\sigma_2^2+d_2^2\right)}{n_1+n_2}$$

$$\Rightarrow \sigma^2=\frac{20\times1+40\times2^2}{60}=3$$

Q88

Let $P(\alpha, \beta, \gamma)$, $Q(0,0, \gamma)$ and $R(\alpha, \beta, -\gamma)$

Now, $\overrightarrow{PQ} \parallel \left(\hat{i}+\hat{j}\right) \Rightarrow -\left(\alpha \hat{i}+\beta \hat{j}\right) \parallel \left(\hat{i}+\hat{j}\right)$
$\Rightarrow \alpha=\beta$

Also, mid point of PQ lies on the plane

$\Rightarrow \frac{\alpha}{2}+\frac{\beta}{2}=3 \Rightarrow \alpha+\beta=6 \Rightarrow \alpha=3$

Now, distance of point P from x-axis is $\sqrt{\beta^2+\gamma^2}=5$

$\Rightarrow \beta^2+\gamma^2=25 \Rightarrow \gamma^2=16$

As $\beta=\alpha=3$

As $\gamma=4$

Hence, $PR=2\gamma=8$

Q89

$\sum_{r=0}^{25}(4r+1)^{25}C_r$

$=4\sum_{r=0}^{25} r.^{25}C_r+\sum_{r=0}^{25}{}^{25}C_r$
$=4\sum_{r=1}^{25} r \times \frac{25}{r}{}^{24}C_{r-1}+2^{25}$

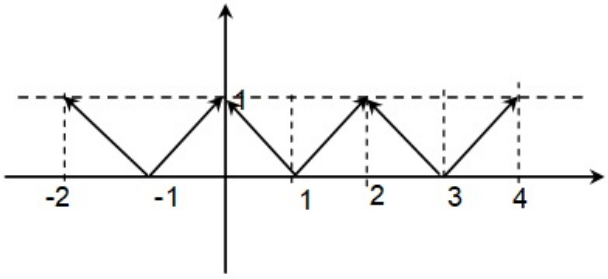
$=100\sum_{r=1}^{25}{}^{25}C_{r-1}+2^{25}$
$=100.2^{24}+2^{25}$

$=2^{25}(50+1)$

$= 51 \ .2^{25}$

So, $k = 51$

Q90



We know that,

$x - [x] = \{x\}$

$\Rightarrow x - [x + 1] = \{x\} - 1$

Hence, $\int_{-2}^{4} f(x)\ dx = 6.\ \frac{1}{2}(1.1) = 3$